

Batch 1

Initial Conditions:

Batch: 1.00

X0 (g/L): 2.00

S0 (g/L): 78.00

M0 (g/L): 42.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.54

F_vol (L/h): 0.03

S_feed (g/L): 195.00

M_feed (g/L): 105.00

V_final (L): 6.88

Batch 1 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.29
Glucose S (g/L): 78.14
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 41.57
Volume V (L): 5.00

t = 1.00 h

Biomass X (g/L): 2.36
Glucose S (g/L): 78.19
Lactic Acid P (g/L): 1.09
Maltose M (g/L): 41.91
Volume V (L): 5.03

t = 2.00 h

Biomass X (g/L): 5.36
Glucose S (g/L): 79.22
Lactic Acid P (g/L): 1.94
Maltose M (g/L): 40.59
Volume V (L): 5.12

t = 3.00 h

Biomass X (g/L): 8.88
Glucose S (g/L): 76.02
Lactic Acid P (g/L): 0.39
Maltose M (g/L): 40.67
Volume V (L): 5.04

t = 4.00 h

Biomass X (g/L): 5.96
Glucose S (g/L): 78.53
Lactic Acid P (g/L): 7.89
Maltose M (g/L): 40.11
Volume V (L): 5.13

t = 5.00 h

Biomass X (g/L): 8.02
Glucose S (g/L): 78.70

Lactic Acid P (g/L): 5.98
Maltose M (g/L): 39.24
Volume V (L): 5.11

t = 6.00 h

Biomass X (g/L): 15.26
Glucose S (g/L): 79.23
Lactic Acid P (g/L): 3.74
Maltose M (g/L): 40.45
Volume V (L): 5.14

t = 7.00 h

Biomass X (g/L): 16.21
Glucose S (g/L): 74.29
Lactic Acid P (g/L): 10.49
Maltose M (g/L): 39.12
Volume V (L): 5.24

t = 8.00 h

Biomass X (g/L): 16.55
Glucose S (g/L): 71.53
Lactic Acid P (g/L): 9.07
Maltose M (g/L): 37.11
Volume V (L): 5.25

t = 9.00 h

Biomass X (g/L): 23.15
Glucose S (g/L): 69.27
Lactic Acid P (g/L): 16.30
Maltose M (g/L): 38.00
Volume V (L): 5.27

t = 10.00 h

Biomass X (g/L): 24.84
Glucose S (g/L): 68.41
Lactic Acid P (g/L): 20.36
Maltose M (g/L): 37.09
Volume V (L): 5.32

t = 11.00 h

Biomass X (g/L): 29.33

Glucose S (g/L): 64.14
Lactic Acid P (g/L): 18.81
Maltose M (g/L): 37.59
Volume V (L): 5.28

t = 12.00 h

Biomass X (g/L): 35.72
Glucose S (g/L): 59.33
Lactic Acid P (g/L): 26.56
Maltose M (g/L): 36.21
Volume V (L): 5.34

t = 13.00 h

Biomass X (g/L): 34.63
Glucose S (g/L): 57.55
Lactic Acid P (g/L): 28.56
Maltose M (g/L): 36.23
Volume V (L): 5.31

t = 14.00 h

Biomass X (g/L): 39.45
Glucose S (g/L): 53.58
Lactic Acid P (g/L): 32.82
Maltose M (g/L): 36.22
Volume V (L): 5.37

t = 15.00 h

Biomass X (g/L): 46.58
Glucose S (g/L): 51.82
Lactic Acid P (g/L): 38.57
Maltose M (g/L): 36.06
Volume V (L): 5.37

t = 16.00 h

Biomass X (g/L): 49.24
Glucose S (g/L): 46.26
Lactic Acid P (g/L): 36.26
Maltose M (g/L): 34.47
Volume V (L): 5.41

t = 17.00 h

Biomass X (g/L): 56.23
Glucose S (g/L): 44.19
Lactic Acid P (g/L): 37.76
Maltose M (g/L): 34.63
Volume V (L): 5.46

t = 18.00 h

Biomass X (g/L): 56.31
Glucose S (g/L): 41.68
Lactic Acid P (g/L): 39.98
Maltose M (g/L): 34.20
Volume V (L): 5.40

t = 19.00 h

Biomass X (g/L): 57.96
Glucose S (g/L): 37.81
Lactic Acid P (g/L): 42.53
Maltose M (g/L): 33.75
Volume V (L): 5.59

t = 20.00 h

Biomass X (g/L): 68.11
Glucose S (g/L): 38.67
Lactic Acid P (g/L): 46.53
Maltose M (g/L): 34.85
Volume V (L): 5.44

t = 21.00 h

Biomass X (g/L): 66.17
Glucose S (g/L): 36.90
Lactic Acid P (g/L): 49.58
Maltose M (g/L): 33.71
Volume V (L): 5.46

t = 22.00 h

Biomass X (g/L): 69.13
Glucose S (g/L): 34.98
Lactic Acid P (g/L): 51.29
Maltose M (g/L): 32.40
Volume V (L): 5.62

t = 23.00 h

Biomass X (g/L): 67.26
Glucose S (g/L): 33.27
Lactic Acid P (g/L): 54.37
Maltose M (g/L): 33.37
Volume V (L): 5.62

t = 24.00 h

Biomass X (g/L): 71.35
Glucose S (g/L): 30.22
Lactic Acid P (g/L): 53.97
Maltose M (g/L): 33.78
Volume V (L): 5.64

t = 25.00 h

Biomass X (g/L): 74.68
Glucose S (g/L): 30.78
Lactic Acid P (g/L): 58.99
Maltose M (g/L): 32.83
Volume V (L): 5.66

t = 26.00 h

Biomass X (g/L): 72.88
Glucose S (g/L): 30.01
Lactic Acid P (g/L): 56.15
Maltose M (g/L): 31.01
Volume V (L): 5.65

t = 27.00 h

Biomass X (g/L): 78.17
Glucose S (g/L): 28.50
Lactic Acid P (g/L): 64.76
Maltose M (g/L): 31.33
Volume V (L): 5.62

t = 28.00 h

Biomass X (g/L): 76.85
Glucose S (g/L): 28.85
Lactic Acid P (g/L): 60.82
Maltose M (g/L): 32.08
Volume V (L): 5.70

t = 29.00 h

Biomass X (g/L): 78.75
Glucose S (g/L): 29.17
Lactic Acid P (g/L): 58.31
Maltose M (g/L): 30.63
Volume V (L): 5.69

t = 30.00 h

Biomass X (g/L): 78.93
Glucose S (g/L): 31.03
Lactic Acid P (g/L): 58.84
Maltose M (g/L): 31.03
Volume V (L): 5.80

t = 31.00 h

Biomass X (g/L): 86.19
Glucose S (g/L): 27.90
Lactic Acid P (g/L): 63.64
Maltose M (g/L): 30.96
Volume V (L): 5.77

t = 32.00 h

Biomass X (g/L): 82.17
Glucose S (g/L): 27.69
Lactic Acid P (g/L): 62.84
Maltose M (g/L): 29.68
Volume V (L): 5.75

t = 33.00 h

Biomass X (g/L): 80.19
Glucose S (g/L): 26.87
Lactic Acid P (g/L): 66.02
Maltose M (g/L): 29.93
Volume V (L): 5.81

t = 34.00 h

Biomass X (g/L): 85.73
Glucose S (g/L): 23.71
Lactic Acid P (g/L): 66.27
Maltose M (g/L): 27.78

Volume V (L): 5.87

t = 35.00 h

Biomass X (g/L): 81.04

Glucose S (g/L): 26.54

Lactic Acid P (g/L): 65.72

Maltose M (g/L): 28.85

Volume V (L): 5.84

t = 36.00 h

Biomass X (g/L): 85.28

Glucose S (g/L): 26.54

Lactic Acid P (g/L): 64.57

Maltose M (g/L): 29.06

Volume V (L): 5.85

t = 37.00 h

Biomass X (g/L): 80.15

Glucose S (g/L): 30.26

Lactic Acid P (g/L): 63.65

Maltose M (g/L): 28.22

Volume V (L): 5.94

t = 38.00 h

Biomass X (g/L): 82.23

Glucose S (g/L): 25.98

Lactic Acid P (g/L): 66.95

Maltose M (g/L): 29.69

Volume V (L): 5.96

t = 39.00 h

Biomass X (g/L): 86.58

Glucose S (g/L): 26.74

Lactic Acid P (g/L): 70.79

Maltose M (g/L): 27.64

Volume V (L): 5.95

t = 40.00 h

Biomass X (g/L): 88.34

Glucose S (g/L): 26.21

Lactic Acid P (g/L): 69.83

Maltose M (g/L): 28.00

Volume V (L): 5.97

t = 41.00 h

Biomass X (g/L): 87.18

Glucose S (g/L): 24.44

Lactic Acid P (g/L): 66.85

Maltose M (g/L): 28.11

Volume V (L): 6.04

t = 42.00 h

Biomass X (g/L): 86.72

Glucose S (g/L): 28.17

Lactic Acid P (g/L): 70.90

Maltose M (g/L): 28.66

Volume V (L): 5.97

t = 43.00 h

Biomass X (g/L): 86.49

Glucose S (g/L): 27.63

Lactic Acid P (g/L): 71.96

Maltose M (g/L): 26.73

Volume V (L): 6.00

t = 44.00 h

Biomass X (g/L): 83.65

Glucose S (g/L): 27.79

Lactic Acid P (g/L): 69.37

Maltose M (g/L): 28.06

Volume V (L): 6.06

t = 45.00 h

Biomass X (g/L): 85.81

Glucose S (g/L): 25.21

Lactic Acid P (g/L): 72.42

Maltose M (g/L): 27.17

Volume V (L): 6.11

t = 46.00 h

Biomass X (g/L): 86.65

Glucose S (g/L): 29.02

Lactic Acid P (g/L): 72.67

Maltose M (g/L): 26.37

Volume V (L): 6.17

t = 47.00 h

Biomass X (g/L): 90.72

Glucose S (g/L): 24.72

Lactic Acid P (g/L): 70.20

Maltose M (g/L): 27.02

Volume V (L): 6.26

t = 48.00 h

Biomass X (g/L): 88.99

Glucose S (g/L): 28.04

Lactic Acid P (g/L): 74.33

Maltose M (g/L): 26.67

Volume V (L): 6.25

t = 49.00 h

Biomass X (g/L): 83.63

Glucose S (g/L): 30.77

Lactic Acid P (g/L): 75.26

Maltose M (g/L): 25.99

Volume V (L): 6.22

t = 50.00 h

Biomass X (g/L): 89.11

Glucose S (g/L): 25.91

Lactic Acid P (g/L): 76.96

Maltose M (g/L): 26.20

Volume V (L): 6.25

t = 51.00 h

Biomass X (g/L): 87.33

Glucose S (g/L): 26.79

Lactic Acid P (g/L): 77.30

Maltose M (g/L): 25.73

Volume V (L): 6.22

t = 52.00 h

Biomass X (g/L): 86.62

Glucose S (g/L): 28.06
Lactic Acid P (g/L): 71.73
Maltose M (g/L): 25.84
Volume V (L): 6.30

t = 53.00 h

Biomass X (g/L): 89.99
Glucose S (g/L): 27.33
Lactic Acid P (g/L): 73.19
Maltose M (g/L): 25.98
Volume V (L): 6.31

t = 54.00 h

Biomass X (g/L): 91.09
Glucose S (g/L): 25.90
Lactic Acid P (g/L): 77.12
Maltose M (g/L): 26.35
Volume V (L): 6.37

t = 55.00 h

Biomass X (g/L): 90.83
Glucose S (g/L): 28.71
Lactic Acid P (g/L): 77.48
Maltose M (g/L): 24.49
Volume V (L): 6.33

t = 56.00 h

Biomass X (g/L): 86.22
Glucose S (g/L): 27.16
Lactic Acid P (g/L): 77.83
Maltose M (g/L): 26.32
Volume V (L): 6.43

t = 57.00 h

Biomass X (g/L): 87.57
Glucose S (g/L): 29.85
Lactic Acid P (g/L): 86.35
Maltose M (g/L): 23.72
Volume V (L): 6.51

t = 58.00 h

Biomass X (g/L): 89.19
Glucose S (g/L): 27.90
Lactic Acid P (g/L): 78.64
Maltose M (g/L): 24.62
Volume V (L): 6.44

t = 59.00 h

Biomass X (g/L): 90.81
Glucose S (g/L): 32.08
Lactic Acid P (g/L): 80.34
Maltose M (g/L): 24.89
Volume V (L): 6.50

t = 60.00 h

Biomass X (g/L): 86.98
Glucose S (g/L): 28.65
Lactic Acid P (g/L): 80.21
Maltose M (g/L): 24.55
Volume V (L): 6.54

t = 61.00 h

Biomass X (g/L): 87.67
Glucose S (g/L): 29.66
Lactic Acid P (g/L): 79.77
Maltose M (g/L): 23.85
Volume V (L): 6.50

t = 62.00 h

Biomass X (g/L): 85.20
Glucose S (g/L): 31.74
Lactic Acid P (g/L): 77.69
Maltose M (g/L): 23.93
Volume V (L): 6.56

t = 63.00 h

Biomass X (g/L): 84.88
Glucose S (g/L): 28.78
Lactic Acid P (g/L): 80.62
Maltose M (g/L): 23.61
Volume V (L): 6.58

t = 64.00 h

Biomass X (g/L): 89.99
Glucose S (g/L): 31.38
Lactic Acid P (g/L): 77.14
Maltose M (g/L): 23.39
Volume V (L): 6.61

t = 65.00 h

Biomass X (g/L): 91.30
Glucose S (g/L): 33.38
Lactic Acid P (g/L): 78.73
Maltose M (g/L): 24.10
Volume V (L): 6.58

t = 66.00 h

Biomass X (g/L): 87.49
Glucose S (g/L): 29.05
Lactic Acid P (g/L): 78.39
Maltose M (g/L): 23.65
Volume V (L): 6.65

t = 67.00 h

Biomass X (g/L): 90.16
Glucose S (g/L): 32.18
Lactic Acid P (g/L): 80.04
Maltose M (g/L): 22.88
Volume V (L): 6.70

t = 68.00 h

Biomass X (g/L): 88.37
Glucose S (g/L): 32.60
Lactic Acid P (g/L): 85.77
Maltose M (g/L): 23.68
Volume V (L): 6.78

t = 69.00 h

Biomass X (g/L): 85.63
Glucose S (g/L): 33.73
Lactic Acid P (g/L): 75.77
Maltose M (g/L): 23.19
Volume V (L): 6.78

t = 70.00 h

Biomass X (g/L): 88.11
Glucose S (g/L): 30.82
Lactic Acid P (g/L): 82.27
Maltose M (g/L): 23.35
Volume V (L): 6.87

t = 71.00 h

Biomass X (g/L): 91.02
Glucose S (g/L): 30.99
Lactic Acid P (g/L): 76.87
Maltose M (g/L): 23.10
Volume V (L): 6.73

t = 72.00 h

Biomass X (g/L): 86.80
Glucose S (g/L): 34.22
Lactic Acid P (g/L): 79.91
Maltose M (g/L): 22.10
Volume V (L): 6.85

t = 73.00 h

Biomass X (g/L): 90.80
Glucose S (g/L): 34.17
Lactic Acid P (g/L): 83.96
Maltose M (g/L): 22.12
Volume V (L): 6.84

t = 74.00 h

Biomass X (g/L): 79.80
Glucose S (g/L): 34.40
Lactic Acid P (g/L): 81.67
Maltose M (g/L): 22.77
Volume V (L): 6.97

t = 75.00 h

Biomass X (g/L): 88.56
Glucose S (g/L): 34.86
Lactic Acid P (g/L): 79.09
Maltose M (g/L): 22.55

Volume V (L): 6.83

Batch 2

Initial Conditions:

Batch: 2.00

X0 (g/L): 3.00

S0 (g/L): 70.00

M0 (g/L): 70.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.00

F_vol (L/h): 0.04

S_feed (g/L): 150.00

M_feed (g/L): 150.00

V_final (L): 8.00

Batch 2 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.88

Glucose S (g/L): 69.87

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 68.39

Volume V (L): 4.89

t = 1.00 h

Biomass X (g/L): 2.54

Glucose S (g/L): 69.70

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 70.64

Volume V (L): 5.10

t = 2.00 h

Biomass X (g/L): 0.07

Glucose S (g/L): 70.82

Lactic Acid P (g/L): 0.24

Maltose M (g/L): 69.18

Volume V (L): 5.06

t = 3.00 h

Biomass X (g/L): 5.88

Glucose S (g/L): 70.19

Lactic Acid P (g/L): 3.19

Maltose M (g/L): 67.85

Volume V (L): 5.09

t = 4.00 h

Biomass X (g/L): 7.52

Glucose S (g/L): 73.55

Lactic Acid P (g/L): 2.67

Maltose M (g/L): 67.49

Volume V (L): 5.22

t = 5.00 h

Biomass X (g/L): 12.54

Glucose S (g/L): 70.39

Lactic Acid P (g/L): 5.79
Maltose M (g/L): 65.56
Volume V (L): 5.24

t = 6.00 h

Biomass X (g/L): 16.27
Glucose S (g/L): 68.09
Lactic Acid P (g/L): 11.45
Maltose M (g/L): 68.15
Volume V (L): 5.21

t = 7.00 h

Biomass X (g/L): 23.86
Glucose S (g/L): 67.20
Lactic Acid P (g/L): 10.42
Maltose M (g/L): 65.41
Volume V (L): 5.38

t = 8.00 h

Biomass X (g/L): 25.64
Glucose S (g/L): 64.42
Lactic Acid P (g/L): 16.78
Maltose M (g/L): 64.78
Volume V (L): 5.22

t = 9.00 h

Biomass X (g/L): 26.18
Glucose S (g/L): 61.57
Lactic Acid P (g/L): 15.36
Maltose M (g/L): 64.72
Volume V (L): 5.42

t = 10.00 h

Biomass X (g/L): 29.92
Glucose S (g/L): 59.63
Lactic Acid P (g/L): 22.65
Maltose M (g/L): 63.90
Volume V (L): 5.45

t = 11.00 h

Biomass X (g/L): 37.97

Glucose S (g/L): 56.82
Lactic Acid P (g/L): 28.19
Maltose M (g/L): 63.09
Volume V (L): 5.41

t = 12.00 h

Biomass X (g/L): 37.85
Glucose S (g/L): 52.37
Lactic Acid P (g/L): 22.24
Maltose M (g/L): 61.59
Volume V (L): 5.51

t = 13.00 h

Biomass X (g/L): 50.05
Glucose S (g/L): 49.52
Lactic Acid P (g/L): 29.42
Maltose M (g/L): 62.17
Volume V (L): 5.41

t = 14.00 h

Biomass X (g/L): 52.40
Glucose S (g/L): 47.28
Lactic Acid P (g/L): 35.72
Maltose M (g/L): 62.89
Volume V (L): 5.64

t = 15.00 h

Biomass X (g/L): 51.95
Glucose S (g/L): 42.53
Lactic Acid P (g/L): 36.73
Maltose M (g/L): 61.86
Volume V (L): 5.58

t = 16.00 h

Biomass X (g/L): 52.22
Glucose S (g/L): 41.34
Lactic Acid P (g/L): 40.76
Maltose M (g/L): 61.55
Volume V (L): 5.59

t = 17.00 h

Biomass X (g/L): 63.06
Glucose S (g/L): 42.85
Lactic Acid P (g/L): 40.91
Maltose M (g/L): 59.12
Volume V (L): 5.77

t = 18.00 h

Biomass X (g/L): 62.19
Glucose S (g/L): 41.54
Lactic Acid P (g/L): 44.83
Maltose M (g/L): 58.17
Volume V (L): 5.66

t = 19.00 h

Biomass X (g/L): 68.16
Glucose S (g/L): 37.90
Lactic Acid P (g/L): 46.36
Maltose M (g/L): 58.44
Volume V (L): 5.63

t = 20.00 h

Biomass X (g/L): 63.33
Glucose S (g/L): 37.65
Lactic Acid P (g/L): 51.43
Maltose M (g/L): 58.11
Volume V (L): 5.66

t = 21.00 h

Biomass X (g/L): 67.93
Glucose S (g/L): 36.32
Lactic Acid P (g/L): 51.05
Maltose M (g/L): 58.56
Volume V (L): 5.89

t = 22.00 h

Biomass X (g/L): 71.34
Glucose S (g/L): 38.71
Lactic Acid P (g/L): 52.90
Maltose M (g/L): 55.56
Volume V (L): 5.76

t = 23.00 h

Biomass X (g/L): 73.14
Glucose S (g/L): 35.70
Lactic Acid P (g/L): 52.65
Maltose M (g/L): 58.00
Volume V (L): 6.08

t = 24.00 h

Biomass X (g/L): 73.42
Glucose S (g/L): 33.66
Lactic Acid P (g/L): 53.88
Maltose M (g/L): 56.01
Volume V (L): 5.77

t = 25.00 h

Biomass X (g/L): 77.52
Glucose S (g/L): 32.23
Lactic Acid P (g/L): 55.32
Maltose M (g/L): 55.31
Volume V (L): 6.15

t = 26.00 h

Biomass X (g/L): 74.50
Glucose S (g/L): 31.11
Lactic Acid P (g/L): 58.50
Maltose M (g/L): 54.33
Volume V (L): 6.06

t = 27.00 h

Biomass X (g/L): 77.96
Glucose S (g/L): 32.99
Lactic Acid P (g/L): 57.71
Maltose M (g/L): 53.30
Volume V (L): 6.07

t = 28.00 h

Biomass X (g/L): 79.64
Glucose S (g/L): 31.69
Lactic Acid P (g/L): 60.46
Maltose M (g/L): 55.11
Volume V (L): 6.07

t = 29.00 h

Biomass X (g/L): 81.55
Glucose S (g/L): 30.81
Lactic Acid P (g/L): 65.69
Maltose M (g/L): 54.01
Volume V (L): 6.20

t = 30.00 h

Biomass X (g/L): 82.88
Glucose S (g/L): 31.69
Lactic Acid P (g/L): 63.77
Maltose M (g/L): 52.35
Volume V (L): 6.20

t = 31.00 h

Biomass X (g/L): 79.00
Glucose S (g/L): 31.21
Lactic Acid P (g/L): 61.81
Maltose M (g/L): 51.93
Volume V (L): 6.34

t = 32.00 h

Biomass X (g/L): 78.64
Glucose S (g/L): 34.55
Lactic Acid P (g/L): 66.26
Maltose M (g/L): 52.39
Volume V (L): 6.29

t = 33.00 h

Biomass X (g/L): 86.33
Glucose S (g/L): 33.73
Lactic Acid P (g/L): 63.22
Maltose M (g/L): 53.81
Volume V (L): 6.33

t = 34.00 h

Biomass X (g/L): 84.49
Glucose S (g/L): 32.86
Lactic Acid P (g/L): 60.10
Maltose M (g/L): 51.67

Volume V (L): 6.33

t = 35.00 h

Biomass X (g/L): 82.25

Glucose S (g/L): 34.82

Lactic Acid P (g/L): 63.24

Maltose M (g/L): 50.15

Volume V (L): 6.39

t = 36.00 h

Biomass X (g/L): 88.49

Glucose S (g/L): 33.42

Lactic Acid P (g/L): 61.86

Maltose M (g/L): 50.91

Volume V (L): 6.47

t = 37.00 h

Biomass X (g/L): 85.26

Glucose S (g/L): 32.59

Lactic Acid P (g/L): 66.09

Maltose M (g/L): 50.51

Volume V (L): 6.33

t = 38.00 h

Biomass X (g/L): 88.29

Glucose S (g/L): 35.69

Lactic Acid P (g/L): 67.57

Maltose M (g/L): 47.94

Volume V (L): 6.40

t = 39.00 h

Biomass X (g/L): 85.79

Glucose S (g/L): 34.86

Lactic Acid P (g/L): 72.07

Maltose M (g/L): 49.97

Volume V (L): 6.63

t = 40.00 h

Biomass X (g/L): 91.08

Glucose S (g/L): 33.36

Lactic Acid P (g/L): 69.41

Maltose M (g/L): 49.45

Volume V (L): 6.62

t = 41.00 h

Biomass X (g/L): 90.55

Glucose S (g/L): 34.72

Lactic Acid P (g/L): 68.64

Maltose M (g/L): 49.94

Volume V (L): 6.62

t = 42.00 h

Biomass X (g/L): 85.70

Glucose S (g/L): 34.31

Lactic Acid P (g/L): 71.62

Maltose M (g/L): 50.63

Volume V (L): 6.68

t = 43.00 h

Biomass X (g/L): 88.95

Glucose S (g/L): 34.11

Lactic Acid P (g/L): 64.88

Maltose M (g/L): 49.64

Volume V (L): 6.75

t = 44.00 h

Biomass X (g/L): 88.28

Glucose S (g/L): 37.24

Lactic Acid P (g/L): 71.14

Maltose M (g/L): 48.03

Volume V (L): 6.71

t = 45.00 h

Biomass X (g/L): 90.22

Glucose S (g/L): 38.82

Lactic Acid P (g/L): 72.85

Maltose M (g/L): 46.94

Volume V (L): 6.73

t = 46.00 h

Biomass X (g/L): 84.44

Glucose S (g/L): 35.38

Lactic Acid P (g/L): 67.93

Maltose M (g/L): 49.95

Volume V (L): 6.86

t = 47.00 h

Biomass X (g/L): 88.68

Glucose S (g/L): 38.13

Lactic Acid P (g/L): 74.55

Maltose M (g/L): 47.34

Volume V (L): 6.79

t = 48.00 h

Biomass X (g/L): 89.68

Glucose S (g/L): 37.16

Lactic Acid P (g/L): 73.03

Maltose M (g/L): 46.98

Volume V (L): 6.96

t = 49.00 h

Biomass X (g/L): 82.61

Glucose S (g/L): 37.82

Lactic Acid P (g/L): 71.61

Maltose M (g/L): 46.63

Volume V (L): 6.81

t = 50.00 h

Biomass X (g/L): 84.08

Glucose S (g/L): 38.17

Lactic Acid P (g/L): 74.46

Maltose M (g/L): 46.53

Volume V (L): 7.09

t = 51.00 h

Biomass X (g/L): 81.93

Glucose S (g/L): 38.33

Lactic Acid P (g/L): 78.69

Maltose M (g/L): 45.91

Volume V (L): 7.08

t = 52.00 h

Biomass X (g/L): 86.39

Glucose S (g/L): 39.89
Lactic Acid P (g/L): 74.07
Maltose M (g/L): 45.22
Volume V (L): 7.10

t = 53.00 h

Biomass X (g/L): 88.85
Glucose S (g/L): 39.34
Lactic Acid P (g/L): 74.55
Maltose M (g/L): 44.95
Volume V (L): 7.21

t = 54.00 h

Biomass X (g/L): 90.80
Glucose S (g/L): 41.13
Lactic Acid P (g/L): 73.18
Maltose M (g/L): 45.12
Volume V (L): 7.31

t = 55.00 h

Biomass X (g/L): 87.15
Glucose S (g/L): 41.25
Lactic Acid P (g/L): 72.56
Maltose M (g/L): 44.37
Volume V (L): 7.29

t = 56.00 h

Biomass X (g/L): 91.01
Glucose S (g/L): 41.52
Lactic Acid P (g/L): 76.83
Maltose M (g/L): 43.93
Volume V (L): 7.07

t = 57.00 h

Biomass X (g/L): 83.35
Glucose S (g/L): 41.24
Lactic Acid P (g/L): 73.12
Maltose M (g/L): 44.39
Volume V (L): 7.16

t = 58.00 h

Biomass X (g/L): 82.45
Glucose S (g/L): 42.13
Lactic Acid P (g/L): 75.59
Maltose M (g/L): 43.78
Volume V (L): 7.26

t = 59.00 h

Biomass X (g/L): 86.52
Glucose S (g/L): 44.20
Lactic Acid P (g/L): 74.55
Maltose M (g/L): 45.10
Volume V (L): 7.36

t = 60.00 h

Biomass X (g/L): 87.53
Glucose S (g/L): 44.40
Lactic Acid P (g/L): 77.08
Maltose M (g/L): 41.07
Volume V (L): 7.45

t = 61.00 h

Biomass X (g/L): 86.37
Glucose S (g/L): 43.17
Lactic Acid P (g/L): 76.98
Maltose M (g/L): 44.19
Volume V (L): 7.37

t = 62.00 h

Biomass X (g/L): 81.12
Glucose S (g/L): 44.94
Lactic Acid P (g/L): 78.89
Maltose M (g/L): 44.07
Volume V (L): 7.50

t = 63.00 h

Biomass X (g/L): 85.99
Glucose S (g/L): 46.21
Lactic Acid P (g/L): 75.46
Maltose M (g/L): 40.80
Volume V (L): 7.45

t = 64.00 h

Biomass X (g/L): 82.80
Glucose S (g/L): 43.87
Lactic Acid P (g/L): 76.25
Maltose M (g/L): 42.11
Volume V (L): 7.50

t = 65.00 h

Biomass X (g/L): 87.64
Glucose S (g/L): 46.98
Lactic Acid P (g/L): 74.79
Maltose M (g/L): 41.83
Volume V (L): 7.47

t = 66.00 h

Biomass X (g/L): 86.73
Glucose S (g/L): 46.07
Lactic Acid P (g/L): 76.28
Maltose M (g/L): 40.64
Volume V (L): 7.56

t = 67.00 h

Biomass X (g/L): 83.29
Glucose S (g/L): 48.02
Lactic Acid P (g/L): 78.44
Maltose M (g/L): 40.97
Volume V (L): 7.56

t = 68.00 h

Biomass X (g/L): 84.21
Glucose S (g/L): 46.95
Lactic Acid P (g/L): 79.54
Maltose M (g/L): 40.42
Volume V (L): 7.63

t = 69.00 h

Biomass X (g/L): 82.68
Glucose S (g/L): 46.23
Lactic Acid P (g/L): 75.76
Maltose M (g/L): 42.77
Volume V (L): 7.85

t = 70.00 h

Biomass X (g/L): 85.03
Glucose S (g/L): 46.94
Lactic Acid P (g/L): 80.20
Maltose M (g/L): 41.80
Volume V (L): 7.71

t = 71.00 h

Biomass X (g/L): 87.44
Glucose S (g/L): 49.40
Lactic Acid P (g/L): 81.54
Maltose M (g/L): 41.87
Volume V (L): 8.08

t = 72.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 50.15
Lactic Acid P (g/L): 79.49
Maltose M (g/L): 41.14
Volume V (L): 7.92

t = 73.00 h

Biomass X (g/L): 85.96
Glucose S (g/L): 49.27
Lactic Acid P (g/L): 83.14
Maltose M (g/L): 39.23
Volume V (L): 7.94

t = 74.00 h

Biomass X (g/L): 83.17
Glucose S (g/L): 51.58
Lactic Acid P (g/L): 77.03
Maltose M (g/L): 39.55
Volume V (L): 7.88

t = 75.00 h

Biomass X (g/L): 82.33
Glucose S (g/L): 49.37
Lactic Acid P (g/L): 76.09
Maltose M (g/L): 40.25

Volume V (L): 8.06

Batch 3

Initial Conditions:

Batch: 3.00

X0 (g/L): 2.50

S0 (g/L): 40.00

M0 (g/L): 60.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.50

F_vol (L/h): 0.06

S_feed (g/L): 120.00

M_feed (g/L): 180.00

V_final (L): 9.12

Batch 3 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 1.12
Glucose S (g/L): 41.37
Lactic Acid P (g/L): 0.56
Maltose M (g/L): 59.93
Volume V (L): 4.96

t = 1.00 h

Biomass X (g/L): 3.60
Glucose S (g/L): 40.31
Lactic Acid P (g/L): 4.16
Maltose M (g/L): 60.15
Volume V (L): 4.92

t = 2.00 h

Biomass X (g/L): 10.50
Glucose S (g/L): 43.00
Lactic Acid P (g/L): 1.17
Maltose M (g/L): 60.34
Volume V (L): 4.95

t = 3.00 h

Biomass X (g/L): 5.48
Glucose S (g/L): 42.54
Lactic Acid P (g/L): 3.01
Maltose M (g/L): 59.75
Volume V (L): 5.31

t = 4.00 h

Biomass X (g/L): 10.17
Glucose S (g/L): 42.92
Lactic Acid P (g/L): 4.99
Maltose M (g/L): 60.40
Volume V (L): 5.16

t = 5.00 h

Biomass X (g/L): 7.82
Glucose S (g/L): 43.37

Lactic Acid P (g/L): 5.58
Maltose M (g/L): 60.34
Volume V (L): 5.32

t = 6.00 h

Biomass X (g/L): 11.94
Glucose S (g/L): 41.13
Lactic Acid P (g/L): 6.00
Maltose M (g/L): 60.30
Volume V (L): 5.32

t = 7.00 h

Biomass X (g/L): 19.22
Glucose S (g/L): 41.36
Lactic Acid P (g/L): 9.60
Maltose M (g/L): 60.57
Volume V (L): 5.44

t = 8.00 h

Biomass X (g/L): 16.82
Glucose S (g/L): 39.63
Lactic Acid P (g/L): 14.04
Maltose M (g/L): 61.24
Volume V (L): 5.45

t = 9.00 h

Biomass X (g/L): 26.35
Glucose S (g/L): 38.01
Lactic Acid P (g/L): 12.01
Maltose M (g/L): 60.91
Volume V (L): 5.34

t = 10.00 h

Biomass X (g/L): 27.62
Glucose S (g/L): 35.65
Lactic Acid P (g/L): 17.51
Maltose M (g/L): 60.55
Volume V (L): 5.67

t = 11.00 h

Biomass X (g/L): 28.61

Glucose S (g/L): 35.22
Lactic Acid P (g/L): 18.62
Maltose M (g/L): 60.89
Volume V (L): 5.54

t = 12.00 h

Biomass X (g/L): 35.53
Glucose S (g/L): 30.07
Lactic Acid P (g/L): 25.91
Maltose M (g/L): 61.13
Volume V (L): 5.47

t = 13.00 h

Biomass X (g/L): 40.31
Glucose S (g/L): 27.80
Lactic Acid P (g/L): 22.76
Maltose M (g/L): 60.96
Volume V (L): 5.66

t = 14.00 h

Biomass X (g/L): 43.27
Glucose S (g/L): 25.46
Lactic Acid P (g/L): 27.80
Maltose M (g/L): 60.66
Volume V (L): 5.96

t = 15.00 h

Biomass X (g/L): 41.59
Glucose S (g/L): 23.43
Lactic Acid P (g/L): 32.65
Maltose M (g/L): 60.40
Volume V (L): 5.93

t = 16.00 h

Biomass X (g/L): 46.95
Glucose S (g/L): 21.48
Lactic Acid P (g/L): 37.90
Maltose M (g/L): 60.76
Volume V (L): 5.84

t = 17.00 h

Biomass X (g/L): 51.17
Glucose S (g/L): 21.43
Lactic Acid P (g/L): 36.62
Maltose M (g/L): 60.60
Volume V (L): 5.89

t = 18.00 h

Biomass X (g/L): 54.39
Glucose S (g/L): 20.18
Lactic Acid P (g/L): 37.75
Maltose M (g/L): 60.68
Volume V (L): 5.95

t = 19.00 h

Biomass X (g/L): 58.61
Glucose S (g/L): 17.19
Lactic Acid P (g/L): 45.29
Maltose M (g/L): 60.52
Volume V (L): 6.30

t = 20.00 h

Biomass X (g/L): 59.86
Glucose S (g/L): 15.91
Lactic Acid P (g/L): 39.69
Maltose M (g/L): 60.56
Volume V (L): 6.15

t = 21.00 h

Biomass X (g/L): 58.33
Glucose S (g/L): 16.21
Lactic Acid P (g/L): 39.76
Maltose M (g/L): 59.98
Volume V (L): 6.21

t = 22.00 h

Biomass X (g/L): 64.35
Glucose S (g/L): 14.61
Lactic Acid P (g/L): 47.32
Maltose M (g/L): 60.45
Volume V (L): 6.34

t = 23.00 h

Biomass X (g/L): 66.62
Glucose S (g/L): 15.11
Lactic Acid P (g/L): 46.70
Maltose M (g/L): 60.00
Volume V (L): 6.29

t = 24.00 h

Biomass X (g/L): 68.07
Glucose S (g/L): 14.31
Lactic Acid P (g/L): 46.93
Maltose M (g/L): 60.46
Volume V (L): 6.29

t = 25.00 h

Biomass X (g/L): 70.74
Glucose S (g/L): 12.48
Lactic Acid P (g/L): 52.12
Maltose M (g/L): 59.50
Volume V (L): 6.35

t = 26.00 h

Biomass X (g/L): 71.44
Glucose S (g/L): 14.99
Lactic Acid P (g/L): 52.29
Maltose M (g/L): 60.12
Volume V (L): 6.42

t = 27.00 h

Biomass X (g/L): 71.73
Glucose S (g/L): 15.09
Lactic Acid P (g/L): 51.61
Maltose M (g/L): 59.51
Volume V (L): 6.48

t = 28.00 h

Biomass X (g/L): 71.54
Glucose S (g/L): 12.91
Lactic Acid P (g/L): 51.07
Maltose M (g/L): 59.12
Volume V (L): 6.63

t = 29.00 h

Biomass X (g/L): 68.71
Glucose S (g/L): 13.14
Lactic Acid P (g/L): 56.93
Maltose M (g/L): 59.03
Volume V (L): 6.60

t = 30.00 h

Biomass X (g/L): 74.64
Glucose S (g/L): 13.82
Lactic Acid P (g/L): 57.38
Maltose M (g/L): 58.94
Volume V (L): 6.74

t = 31.00 h

Biomass X (g/L): 74.94
Glucose S (g/L): 14.00
Lactic Acid P (g/L): 56.59
Maltose M (g/L): 59.02
Volume V (L): 6.70

t = 32.00 h

Biomass X (g/L): 75.86
Glucose S (g/L): 14.48
Lactic Acid P (g/L): 61.83
Maltose M (g/L): 58.89
Volume V (L): 6.77

t = 33.00 h

Biomass X (g/L): 72.86
Glucose S (g/L): 15.90
Lactic Acid P (g/L): 56.08
Maltose M (g/L): 58.40
Volume V (L): 6.57

t = 34.00 h

Biomass X (g/L): 73.98
Glucose S (g/L): 14.23
Lactic Acid P (g/L): 55.83
Maltose M (g/L): 58.41

Volume V (L): 6.98

t = 35.00 h

Biomass X (g/L): 79.70

Glucose S (g/L): 13.98

Lactic Acid P (g/L): 58.45

Maltose M (g/L): 58.91

Volume V (L): 6.97

t = 36.00 h

Biomass X (g/L): 77.63

Glucose S (g/L): 13.79

Lactic Acid P (g/L): 60.61

Maltose M (g/L): 58.62

Volume V (L): 7.10

t = 37.00 h

Biomass X (g/L): 79.86

Glucose S (g/L): 14.72

Lactic Acid P (g/L): 61.94

Maltose M (g/L): 58.05

Volume V (L): 6.68

t = 38.00 h

Biomass X (g/L): 78.77

Glucose S (g/L): 15.72

Lactic Acid P (g/L): 61.49

Maltose M (g/L): 57.76

Volume V (L): 7.35

t = 39.00 h

Biomass X (g/L): 79.20

Glucose S (g/L): 17.75

Lactic Acid P (g/L): 63.45

Maltose M (g/L): 57.61

Volume V (L): 7.13

t = 40.00 h

Biomass X (g/L): 81.77

Glucose S (g/L): 16.12

Lactic Acid P (g/L): 60.44

Maltose M (g/L): 57.58

Volume V (L): 7.34

t = 41.00 h

Biomass X (g/L): 78.64

Glucose S (g/L): 17.22

Lactic Acid P (g/L): 67.08

Maltose M (g/L): 56.90

Volume V (L): 7.13

t = 42.00 h

Biomass X (g/L): 80.44

Glucose S (g/L): 17.47

Lactic Acid P (g/L): 64.19

Maltose M (g/L): 56.95

Volume V (L): 7.39

t = 43.00 h

Biomass X (g/L): 79.40

Glucose S (g/L): 19.02

Lactic Acid P (g/L): 67.42

Maltose M (g/L): 56.34

Volume V (L): 7.23

t = 44.00 h

Biomass X (g/L): 79.74

Glucose S (g/L): 20.69

Lactic Acid P (g/L): 66.19

Maltose M (g/L): 56.79

Volume V (L): 7.34

t = 45.00 h

Biomass X (g/L): 80.29

Glucose S (g/L): 18.49

Lactic Acid P (g/L): 66.93

Maltose M (g/L): 56.67

Volume V (L): 7.71

t = 46.00 h

Biomass X (g/L): 81.78

Glucose S (g/L): 19.04

Lactic Acid P (g/L): 67.66

Maltose M (g/L): 56.14

Volume V (L): 7.51

t = 47.00 h

Biomass X (g/L): 80.30

Glucose S (g/L): 20.93

Lactic Acid P (g/L): 67.84

Maltose M (g/L): 56.58

Volume V (L): 7.61

t = 48.00 h

Biomass X (g/L): 84.64

Glucose S (g/L): 21.14

Lactic Acid P (g/L): 68.72

Maltose M (g/L): 55.53

Volume V (L): 7.75

t = 49.00 h

Biomass X (g/L): 79.64

Glucose S (g/L): 22.72

Lactic Acid P (g/L): 70.69

Maltose M (g/L): 55.26

Volume V (L): 7.76

t = 50.00 h

Biomass X (g/L): 81.47

Glucose S (g/L): 22.16

Lactic Acid P (g/L): 68.55

Maltose M (g/L): 55.58

Volume V (L): 7.77

t = 51.00 h

Biomass X (g/L): 80.99

Glucose S (g/L): 21.88

Lactic Acid P (g/L): 70.09

Maltose M (g/L): 55.64

Volume V (L): 7.85

t = 52.00 h

Biomass X (g/L): 85.61

Glucose S (g/L): 23.29
Lactic Acid P (g/L): 68.68
Maltose M (g/L): 55.64
Volume V (L): 8.16

t = 53.00 h

Biomass X (g/L): 82.70
Glucose S (g/L): 24.32
Lactic Acid P (g/L): 72.49
Maltose M (g/L): 54.79
Volume V (L): 7.91

t = 54.00 h

Biomass X (g/L): 84.78
Glucose S (g/L): 24.65
Lactic Acid P (g/L): 68.08
Maltose M (g/L): 55.06
Volume V (L): 7.99

t = 55.00 h

Biomass X (g/L): 78.79
Glucose S (g/L): 24.95
Lactic Acid P (g/L): 74.00
Maltose M (g/L): 54.52
Volume V (L): 8.16

t = 56.00 h

Biomass X (g/L): 83.04
Glucose S (g/L): 26.04
Lactic Acid P (g/L): 70.19
Maltose M (g/L): 54.28
Volume V (L): 8.22

t = 57.00 h

Biomass X (g/L): 84.56
Glucose S (g/L): 26.72
Lactic Acid P (g/L): 67.39
Maltose M (g/L): 54.40
Volume V (L): 8.28

t = 58.00 h

Biomass X (g/L): 82.64
Glucose S (g/L): 27.50
Lactic Acid P (g/L): 71.21
Maltose M (g/L): 54.13
Volume V (L): 8.27

t = 59.00 h

Biomass X (g/L): 85.00
Glucose S (g/L): 27.45
Lactic Acid P (g/L): 69.69
Maltose M (g/L): 53.24
Volume V (L): 8.10

t = 60.00 h

Biomass X (g/L): 81.20
Glucose S (g/L): 27.33
Lactic Acid P (g/L): 73.41
Maltose M (g/L): 53.98
Volume V (L): 8.50

t = 61.00 h

Biomass X (g/L): 85.81
Glucose S (g/L): 28.21
Lactic Acid P (g/L): 70.33
Maltose M (g/L): 53.75
Volume V (L): 8.21

t = 62.00 h

Biomass X (g/L): 87.91
Glucose S (g/L): 29.75
Lactic Acid P (g/L): 71.08
Maltose M (g/L): 52.93
Volume V (L): 8.45

t = 63.00 h

Biomass X (g/L): 81.49
Glucose S (g/L): 28.56
Lactic Acid P (g/L): 68.10
Maltose M (g/L): 53.04
Volume V (L): 8.37

t = 64.00 h

Biomass X (g/L): 81.25
Glucose S (g/L): 30.21
Lactic Acid P (g/L): 71.59
Maltose M (g/L): 52.87
Volume V (L): 8.51

t = 65.00 h

Biomass X (g/L): 85.75
Glucose S (g/L): 30.15
Lactic Acid P (g/L): 67.41
Maltose M (g/L): 52.59
Volume V (L): 8.62

t = 66.00 h

Biomass X (g/L): 85.99
Glucose S (g/L): 31.12
Lactic Acid P (g/L): 69.55
Maltose M (g/L): 52.14
Volume V (L): 8.67

t = 67.00 h

Biomass X (g/L): 80.88
Glucose S (g/L): 32.84
Lactic Acid P (g/L): 75.07
Maltose M (g/L): 51.90
Volume V (L): 8.74

t = 68.00 h

Biomass X (g/L): 81.05
Glucose S (g/L): 32.48
Lactic Acid P (g/L): 75.36
Maltose M (g/L): 51.90
Volume V (L): 8.94

t = 69.00 h

Biomass X (g/L): 81.30
Glucose S (g/L): 32.96
Lactic Acid P (g/L): 74.77
Maltose M (g/L): 52.06
Volume V (L): 8.85

t = 70.00 h

Biomass X (g/L): 78.67
Glucose S (g/L): 32.32
Lactic Acid P (g/L): 71.86
Maltose M (g/L): 51.67
Volume V (L): 8.82

t = 71.00 h

Biomass X (g/L): 79.59
Glucose S (g/L): 34.78
Lactic Acid P (g/L): 74.14
Maltose M (g/L): 51.19
Volume V (L): 9.02

t = 72.00 h

Biomass X (g/L): 79.29
Glucose S (g/L): 35.30
Lactic Acid P (g/L): 74.45
Maltose M (g/L): 51.34
Volume V (L): 9.11

t = 73.00 h

Biomass X (g/L): 79.75
Glucose S (g/L): 37.24
Lactic Acid P (g/L): 72.06
Maltose M (g/L): 50.02
Volume V (L): 8.86

t = 74.00 h

Biomass X (g/L): 81.40
Glucose S (g/L): 35.65
Lactic Acid P (g/L): 78.25
Maltose M (g/L): 51.05
Volume V (L): 9.14

t = 75.00 h

Biomass X (g/L): 81.92
Glucose S (g/L): 36.72
Lactic Acid P (g/L): 77.03
Maltose M (g/L): 49.96

Volume V (L): 9.21

Batch 4

Initial Conditions:

Batch: 4.00

X0 (g/L): 1.50

S0 (g/L): 78.00

M0 (g/L): 52.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.67

F_vol (L/h): 0.04

S_feed (g/L): 180.00

M_feed (g/L): 120.00

V_final (L): 7.62

Batch 4 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.74

Glucose S (g/L): 77.09

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 51.70

Volume V (L): 5.03

t = 1.00 h

Biomass X (g/L): 5.56

Glucose S (g/L): 79.54

Lactic Acid P (g/L): 2.40

Maltose M (g/L): 50.88

Volume V (L): 4.98

t = 2.00 h

Biomass X (g/L): 2.37

Glucose S (g/L): 79.00

Lactic Acid P (g/L): 1.11

Maltose M (g/L): 50.84

Volume V (L): 5.03

t = 3.00 h

Biomass X (g/L): 4.03

Glucose S (g/L): 83.72

Lactic Acid P (g/L): 1.38

Maltose M (g/L): 51.59

Volume V (L): 5.15

t = 4.00 h

Biomass X (g/L): 4.51

Glucose S (g/L): 79.62

Lactic Acid P (g/L): 4.71

Maltose M (g/L): 49.03

Volume V (L): 5.24

t = 5.00 h

Biomass X (g/L): 6.59

Glucose S (g/L): 81.64

Lactic Acid P (g/L): 2.99
Maltose M (g/L): 48.38
Volume V (L): 5.18

t = 6.00 h

Biomass X (g/L): 5.76
Glucose S (g/L): 81.64
Lactic Acid P (g/L): 5.79
Maltose M (g/L): 49.96
Volume V (L): 5.23

t = 7.00 h

Biomass X (g/L): 7.37
Glucose S (g/L): 80.41
Lactic Acid P (g/L): 7.72
Maltose M (g/L): 48.37
Volume V (L): 5.38

t = 8.00 h

Biomass X (g/L): 18.13
Glucose S (g/L): 78.15
Lactic Acid P (g/L): 7.53
Maltose M (g/L): 47.60
Volume V (L): 5.30

t = 9.00 h

Biomass X (g/L): 15.32
Glucose S (g/L): 76.27
Lactic Acid P (g/L): 14.01
Maltose M (g/L): 48.29
Volume V (L): 5.52

t = 10.00 h

Biomass X (g/L): 18.82
Glucose S (g/L): 74.10
Lactic Acid P (g/L): 14.05
Maltose M (g/L): 46.75
Volume V (L): 5.39

t = 11.00 h

Biomass X (g/L): 20.04

Glucose S (g/L): 71.08
Lactic Acid P (g/L): 15.73
Maltose M (g/L): 47.94
Volume V (L): 5.25

t = 12.00 h

Biomass X (g/L): 28.31
Glucose S (g/L): 69.57
Lactic Acid P (g/L): 22.05
Maltose M (g/L): 46.74
Volume V (L): 5.48

t = 13.00 h

Biomass X (g/L): 31.02
Glucose S (g/L): 68.13
Lactic Acid P (g/L): 21.92
Maltose M (g/L): 45.54
Volume V (L): 5.49

t = 14.00 h

Biomass X (g/L): 43.04
Glucose S (g/L): 61.28
Lactic Acid P (g/L): 27.42
Maltose M (g/L): 46.90
Volume V (L): 5.59

t = 15.00 h

Biomass X (g/L): 45.68
Glucose S (g/L): 57.16
Lactic Acid P (g/L): 30.54
Maltose M (g/L): 44.65
Volume V (L): 5.58

t = 16.00 h

Biomass X (g/L): 45.38
Glucose S (g/L): 56.02
Lactic Acid P (g/L): 36.22
Maltose M (g/L): 45.15
Volume V (L): 5.55

t = 17.00 h

Biomass X (g/L): 56.71
Glucose S (g/L): 56.96
Lactic Acid P (g/L): 37.67
Maltose M (g/L): 44.47
Volume V (L): 5.50

t = 18.00 h

Biomass X (g/L): 56.97
Glucose S (g/L): 49.98
Lactic Acid P (g/L): 39.19
Maltose M (g/L): 44.76
Volume V (L): 5.61

t = 19.00 h

Biomass X (g/L): 56.84
Glucose S (g/L): 51.09
Lactic Acid P (g/L): 43.74
Maltose M (g/L): 44.55
Volume V (L): 5.60

t = 20.00 h

Biomass X (g/L): 54.21
Glucose S (g/L): 49.20
Lactic Acid P (g/L): 47.46
Maltose M (g/L): 43.16
Volume V (L): 5.65

t = 21.00 h

Biomass X (g/L): 69.18
Glucose S (g/L): 44.48
Lactic Acid P (g/L): 47.64
Maltose M (g/L): 42.27
Volume V (L): 5.78

t = 22.00 h

Biomass X (g/L): 62.17
Glucose S (g/L): 44.48
Lactic Acid P (g/L): 48.34
Maltose M (g/L): 42.73
Volume V (L): 5.90

t = 23.00 h

Biomass X (g/L): 63.71
Glucose S (g/L): 44.59
Lactic Acid P (g/L): 47.04
Maltose M (g/L): 41.62
Volume V (L): 5.84

t = 24.00 h

Biomass X (g/L): 72.55
Glucose S (g/L): 40.75
Lactic Acid P (g/L): 49.34
Maltose M (g/L): 43.07
Volume V (L): 5.75

t = 25.00 h

Biomass X (g/L): 77.98
Glucose S (g/L): 40.00
Lactic Acid P (g/L): 52.74
Maltose M (g/L): 41.48
Volume V (L): 5.91

t = 26.00 h

Biomass X (g/L): 76.90
Glucose S (g/L): 38.66
Lactic Acid P (g/L): 53.48
Maltose M (g/L): 40.81
Volume V (L): 6.00

t = 27.00 h

Biomass X (g/L): 76.27
Glucose S (g/L): 37.37
Lactic Acid P (g/L): 55.86
Maltose M (g/L): 40.85
Volume V (L): 5.90

t = 28.00 h

Biomass X (g/L): 77.68
Glucose S (g/L): 37.27
Lactic Acid P (g/L): 57.71
Maltose M (g/L): 39.50
Volume V (L): 5.97

t = 29.00 h

Biomass X (g/L): 79.55
Glucose S (g/L): 36.00
Lactic Acid P (g/L): 60.03
Maltose M (g/L): 39.44
Volume V (L): 6.02

t = 30.00 h

Biomass X (g/L): 80.39
Glucose S (g/L): 37.30
Lactic Acid P (g/L): 59.70
Maltose M (g/L): 40.58
Volume V (L): 5.99

t = 31.00 h

Biomass X (g/L): 80.15
Glucose S (g/L): 35.90
Lactic Acid P (g/L): 62.07
Maltose M (g/L): 39.03
Volume V (L): 6.14

t = 32.00 h

Biomass X (g/L): 80.58
Glucose S (g/L): 37.02
Lactic Acid P (g/L): 62.03
Maltose M (g/L): 38.47
Volume V (L): 6.11

t = 33.00 h

Biomass X (g/L): 80.96
Glucose S (g/L): 34.43
Lactic Acid P (g/L): 58.08
Maltose M (g/L): 39.23
Volume V (L): 6.19

t = 34.00 h

Biomass X (g/L): 79.91
Glucose S (g/L): 36.86
Lactic Acid P (g/L): 61.41
Maltose M (g/L): 39.38

Volume V (L): 6.12

t = 35.00 h

Biomass X (g/L): 81.78

Glucose S (g/L): 37.48

Lactic Acid P (g/L): 64.12

Maltose M (g/L): 38.78

Volume V (L): 6.19

t = 36.00 h

Biomass X (g/L): 78.75

Glucose S (g/L): 33.66

Lactic Acid P (g/L): 62.35

Maltose M (g/L): 38.68

Volume V (L): 6.32

t = 37.00 h

Biomass X (g/L): 83.36

Glucose S (g/L): 36.88

Lactic Acid P (g/L): 67.39

Maltose M (g/L): 37.23

Volume V (L): 6.27

t = 38.00 h

Biomass X (g/L): 81.56

Glucose S (g/L): 38.10

Lactic Acid P (g/L): 70.31

Maltose M (g/L): 36.51

Volume V (L): 6.47

t = 39.00 h

Biomass X (g/L): 81.69

Glucose S (g/L): 34.92

Lactic Acid P (g/L): 70.22

Maltose M (g/L): 38.20

Volume V (L): 6.33

t = 40.00 h

Biomass X (g/L): 85.36

Glucose S (g/L): 38.94

Lactic Acid P (g/L): 67.42

Maltose M (g/L): 36.71

Volume V (L): 6.42

t = 41.00 h

Biomass X (g/L): 88.81

Glucose S (g/L): 37.42

Lactic Acid P (g/L): 72.08

Maltose M (g/L): 36.50

Volume V (L): 6.51

t = 42.00 h

Biomass X (g/L): 87.89

Glucose S (g/L): 36.10

Lactic Acid P (g/L): 69.44

Maltose M (g/L): 37.30

Volume V (L): 6.36

t = 43.00 h

Biomass X (g/L): 81.77

Glucose S (g/L): 37.19

Lactic Acid P (g/L): 68.82

Maltose M (g/L): 36.53

Volume V (L): 6.37

t = 44.00 h

Biomass X (g/L): 83.49

Glucose S (g/L): 38.26

Lactic Acid P (g/L): 68.68

Maltose M (g/L): 36.21

Volume V (L): 6.42

t = 45.00 h

Biomass X (g/L): 88.64

Glucose S (g/L): 37.86

Lactic Acid P (g/L): 69.91

Maltose M (g/L): 37.11

Volume V (L): 6.67

t = 46.00 h

Biomass X (g/L): 83.88

Glucose S (g/L): 38.76

Lactic Acid P (g/L): 70.18

Maltose M (g/L): 34.99

Volume V (L): 6.57

t = 47.00 h

Biomass X (g/L): 85.97

Glucose S (g/L): 37.09

Lactic Acid P (g/L): 71.97

Maltose M (g/L): 35.80

Volume V (L): 6.85

t = 48.00 h

Biomass X (g/L): 88.08

Glucose S (g/L): 38.86

Lactic Acid P (g/L): 72.38

Maltose M (g/L): 34.98

Volume V (L): 6.64

t = 49.00 h

Biomass X (g/L): 84.29

Glucose S (g/L): 37.04

Lactic Acid P (g/L): 72.43

Maltose M (g/L): 34.89

Volume V (L): 6.73

t = 50.00 h

Biomass X (g/L): 87.12

Glucose S (g/L): 39.13

Lactic Acid P (g/L): 73.86

Maltose M (g/L): 35.51

Volume V (L): 6.87

t = 51.00 h

Biomass X (g/L): 83.49

Glucose S (g/L): 38.40

Lactic Acid P (g/L): 76.05

Maltose M (g/L): 34.60

Volume V (L): 6.94

t = 52.00 h

Biomass X (g/L): 85.41

Glucose S (g/L): 38.97
Lactic Acid P (g/L): 75.85
Maltose M (g/L): 34.40
Volume V (L): 6.98

t = 53.00 h

Biomass X (g/L): 87.79
Glucose S (g/L): 41.99
Lactic Acid P (g/L): 70.40
Maltose M (g/L): 34.54
Volume V (L): 6.95

t = 54.00 h

Biomass X (g/L): 82.90
Glucose S (g/L): 42.16
Lactic Acid P (g/L): 68.19
Maltose M (g/L): 33.36
Volume V (L): 6.97

t = 55.00 h

Biomass X (g/L): 88.10
Glucose S (g/L): 41.88
Lactic Acid P (g/L): 76.86
Maltose M (g/L): 32.83
Volume V (L): 6.97

t = 56.00 h

Biomass X (g/L): 86.90
Glucose S (g/L): 39.97
Lactic Acid P (g/L): 71.69
Maltose M (g/L): 32.48
Volume V (L): 7.02

t = 57.00 h

Biomass X (g/L): 88.34
Glucose S (g/L): 41.40
Lactic Acid P (g/L): 74.73
Maltose M (g/L): 31.66
Volume V (L): 6.95

t = 58.00 h

Biomass X (g/L): 87.45
Glucose S (g/L): 43.84
Lactic Acid P (g/L): 72.78
Maltose M (g/L): 34.15
Volume V (L): 6.97

t = 59.00 h

Biomass X (g/L): 90.32
Glucose S (g/L): 42.80
Lactic Acid P (g/L): 71.57
Maltose M (g/L): 31.83
Volume V (L): 7.06

t = 60.00 h

Biomass X (g/L): 87.07
Glucose S (g/L): 46.12
Lactic Acid P (g/L): 77.45
Maltose M (g/L): 32.22
Volume V (L): 7.09

t = 61.00 h

Biomass X (g/L): 85.57
Glucose S (g/L): 43.93
Lactic Acid P (g/L): 78.25
Maltose M (g/L): 31.99
Volume V (L): 7.10

t = 62.00 h

Biomass X (g/L): 86.90
Glucose S (g/L): 44.06
Lactic Acid P (g/L): 75.33
Maltose M (g/L): 32.19
Volume V (L): 7.22

t = 63.00 h

Biomass X (g/L): 87.88
Glucose S (g/L): 43.80
Lactic Acid P (g/L): 70.78
Maltose M (g/L): 32.34
Volume V (L): 7.28

t = 64.00 h

Biomass X (g/L): 86.51
Glucose S (g/L): 44.80
Lactic Acid P (g/L): 75.91
Maltose M (g/L): 32.18
Volume V (L): 7.25

t = 65.00 h

Biomass X (g/L): 86.38
Glucose S (g/L): 44.82
Lactic Acid P (g/L): 78.40
Maltose M (g/L): 31.03
Volume V (L): 7.39

t = 66.00 h

Biomass X (g/L): 84.36
Glucose S (g/L): 46.39
Lactic Acid P (g/L): 74.17
Maltose M (g/L): 31.13
Volume V (L): 7.22

t = 67.00 h

Biomass X (g/L): 84.33
Glucose S (g/L): 47.14
Lactic Acid P (g/L): 78.38
Maltose M (g/L): 31.25
Volume V (L): 7.33

t = 68.00 h

Biomass X (g/L): 82.30
Glucose S (g/L): 45.21
Lactic Acid P (g/L): 78.13
Maltose M (g/L): 30.71
Volume V (L): 7.32

t = 69.00 h

Biomass X (g/L): 90.25
Glucose S (g/L): 44.89
Lactic Acid P (g/L): 79.77
Maltose M (g/L): 31.78
Volume V (L): 7.27

t = 70.00 h

Biomass X (g/L): 82.43
Glucose S (g/L): 44.62
Lactic Acid P (g/L): 78.89
Maltose M (g/L): 30.86
Volume V (L): 7.44

t = 71.00 h

Biomass X (g/L): 83.69
Glucose S (g/L): 48.20
Lactic Acid P (g/L): 76.49
Maltose M (g/L): 30.85
Volume V (L): 7.48

t = 72.00 h

Biomass X (g/L): 83.48
Glucose S (g/L): 46.44
Lactic Acid P (g/L): 81.88
Maltose M (g/L): 30.63
Volume V (L): 7.64

t = 73.00 h

Biomass X (g/L): 87.46
Glucose S (g/L): 45.44
Lactic Acid P (g/L): 78.84
Maltose M (g/L): 30.81
Volume V (L): 7.60

t = 74.00 h

Biomass X (g/L): 84.24
Glucose S (g/L): 49.28
Lactic Acid P (g/L): 79.35
Maltose M (g/L): 29.98
Volume V (L): 7.51

t = 75.00 h

Biomass X (g/L): 88.03
Glucose S (g/L): 52.53
Lactic Acid P (g/L): 77.38
Maltose M (g/L): 28.82

Volume V (L): 7.77

Batch 5

Initial Conditions:

Batch: 5.00

X0 (g/L): 4.00

S0 (g/L): 67.50

M0 (g/L): 82.50

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.22

F_vol (L/h): 0.02

S_feed (g/L): 135.00

M_feed (g/L): 165.00

V_final (L): 6.50

Batch 5 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 7.18
Glucose S (g/L): 68.33
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 82.77
Volume V (L): 5.01

t = 1.00 h

Biomass X (g/L): 6.93
Glucose S (g/L): 70.71
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 82.67
Volume V (L): 5.04

t = 2.00 h

Biomass X (g/L): 6.66
Glucose S (g/L): 67.95
Lactic Acid P (g/L): 6.43
Maltose M (g/L): 76.14
Volume V (L): 4.99

t = 3.00 h

Biomass X (g/L): 7.07
Glucose S (g/L): 66.41
Lactic Acid P (g/L): 3.24
Maltose M (g/L): 79.78
Volume V (L): 5.12

t = 4.00 h

Biomass X (g/L): 11.47
Glucose S (g/L): 66.03
Lactic Acid P (g/L): 9.20
Maltose M (g/L): 76.69
Volume V (L): 5.08

t = 5.00 h

Biomass X (g/L): 18.69
Glucose S (g/L): 66.43

Lactic Acid P (g/L): 12.46

Maltose M (g/L): 73.86

Volume V (L): 5.10

t = 6.00 h

Biomass X (g/L): 24.69

Glucose S (g/L): 61.46

Lactic Acid P (g/L): 14.01

Maltose M (g/L): 76.08

Volume V (L): 5.15

t = 7.00 h

Biomass X (g/L): 28.15

Glucose S (g/L): 58.54

Lactic Acid P (g/L): 13.59

Maltose M (g/L): 76.22

Volume V (L): 5.15

t = 8.00 h

Biomass X (g/L): 27.50

Glucose S (g/L): 55.92

Lactic Acid P (g/L): 19.10

Maltose M (g/L): 72.98

Volume V (L): 5.20

t = 9.00 h

Biomass X (g/L): 30.63

Glucose S (g/L): 52.05

Lactic Acid P (g/L): 21.54

Maltose M (g/L): 73.48

Volume V (L): 5.19

t = 10.00 h

Biomass X (g/L): 38.68

Glucose S (g/L): 50.21

Lactic Acid P (g/L): 22.87

Maltose M (g/L): 73.39

Volume V (L): 5.17

t = 11.00 h

Biomass X (g/L): 38.34

Glucose S (g/L): 49.03
Lactic Acid P (g/L): 29.42
Maltose M (g/L): 71.96
Volume V (L): 5.24

t = 12.00 h

Biomass X (g/L): 48.19
Glucose S (g/L): 44.23
Lactic Acid P (g/L): 28.96
Maltose M (g/L): 71.44
Volume V (L): 5.19

t = 13.00 h

Biomass X (g/L): 51.08
Glucose S (g/L): 40.63
Lactic Acid P (g/L): 38.07
Maltose M (g/L): 69.19
Volume V (L): 5.26

t = 14.00 h

Biomass X (g/L): 54.06
Glucose S (g/L): 38.01
Lactic Acid P (g/L): 36.85
Maltose M (g/L): 67.29
Volume V (L): 5.28

t = 15.00 h

Biomass X (g/L): 54.61
Glucose S (g/L): 33.99
Lactic Acid P (g/L): 41.26
Maltose M (g/L): 63.97
Volume V (L): 5.29

t = 16.00 h

Biomass X (g/L): 63.28
Glucose S (g/L): 35.24
Lactic Acid P (g/L): 43.99
Maltose M (g/L): 65.00
Volume V (L): 5.31

t = 17.00 h

Biomass X (g/L): 63.44
Glucose S (g/L): 32.15
Lactic Acid P (g/L): 44.96
Maltose M (g/L): 63.26
Volume V (L): 5.26

t = 18.00 h

Biomass X (g/L): 64.38
Glucose S (g/L): 30.44
Lactic Acid P (g/L): 49.24
Maltose M (g/L): 66.71
Volume V (L): 5.34

t = 19.00 h

Biomass X (g/L): 62.31
Glucose S (g/L): 33.63
Lactic Acid P (g/L): 48.55
Maltose M (g/L): 62.50
Volume V (L): 5.38

t = 20.00 h

Biomass X (g/L): 71.84
Glucose S (g/L): 28.55
Lactic Acid P (g/L): 53.77
Maltose M (g/L): 62.10
Volume V (L): 5.41

t = 21.00 h

Biomass X (g/L): 78.45
Glucose S (g/L): 26.13
Lactic Acid P (g/L): 49.88
Maltose M (g/L): 61.90
Volume V (L): 5.37

t = 22.00 h

Biomass X (g/L): 79.89
Glucose S (g/L): 25.71
Lactic Acid P (g/L): 55.15
Maltose M (g/L): 60.38
Volume V (L): 5.50

t = 23.00 h

Biomass X (g/L): 75.94
Glucose S (g/L): 27.11
Lactic Acid P (g/L): 61.17
Maltose M (g/L): 61.96
Volume V (L): 5.42

t = 24.00 h

Biomass X (g/L): 76.98
Glucose S (g/L): 25.11
Lactic Acid P (g/L): 55.13
Maltose M (g/L): 57.88
Volume V (L): 5.52

t = 25.00 h

Biomass X (g/L): 81.05
Glucose S (g/L): 22.51
Lactic Acid P (g/L): 57.21
Maltose M (g/L): 58.31
Volume V (L): 5.52

t = 26.00 h

Biomass X (g/L): 78.39
Glucose S (g/L): 25.28
Lactic Acid P (g/L): 61.40
Maltose M (g/L): 56.24
Volume V (L): 5.49

t = 27.00 h

Biomass X (g/L): 82.83
Glucose S (g/L): 22.81
Lactic Acid P (g/L): 61.67
Maltose M (g/L): 54.69
Volume V (L): 5.62

t = 28.00 h

Biomass X (g/L): 85.00
Glucose S (g/L): 25.98
Lactic Acid P (g/L): 63.15
Maltose M (g/L): 55.01
Volume V (L): 5.51

t = 29.00 h

Biomass X (g/L): 87.06
Glucose S (g/L): 24.72
Lactic Acid P (g/L): 64.49
Maltose M (g/L): 55.14
Volume V (L): 5.62

t = 30.00 h

Biomass X (g/L): 84.40
Glucose S (g/L): 22.52
Lactic Acid P (g/L): 65.79
Maltose M (g/L): 53.91
Volume V (L): 5.65

t = 31.00 h

Biomass X (g/L): 84.84
Glucose S (g/L): 24.77
Lactic Acid P (g/L): 67.87
Maltose M (g/L): 52.88
Volume V (L): 5.59

t = 32.00 h

Biomass X (g/L): 83.73
Glucose S (g/L): 23.09
Lactic Acid P (g/L): 66.95
Maltose M (g/L): 51.21
Volume V (L): 5.70

t = 33.00 h

Biomass X (g/L): 84.45
Glucose S (g/L): 24.15
Lactic Acid P (g/L): 66.11
Maltose M (g/L): 52.47
Volume V (L): 5.65

t = 34.00 h

Biomass X (g/L): 86.54
Glucose S (g/L): 23.69
Lactic Acid P (g/L): 69.17
Maltose M (g/L): 49.82

Volume V (L): 5.69

t = 35.00 h

Biomass X (g/L): 88.57

Glucose S (g/L): 21.96

Lactic Acid P (g/L): 72.98

Maltose M (g/L): 52.45

Volume V (L): 5.71

t = 36.00 h

Biomass X (g/L): 90.42

Glucose S (g/L): 22.33

Lactic Acid P (g/L): 68.56

Maltose M (g/L): 50.89

Volume V (L): 5.71

t = 37.00 h

Biomass X (g/L): 86.34

Glucose S (g/L): 24.71

Lactic Acid P (g/L): 67.11

Maltose M (g/L): 48.11

Volume V (L): 5.78

t = 38.00 h

Biomass X (g/L): 92.67

Glucose S (g/L): 23.02

Lactic Acid P (g/L): 72.98

Maltose M (g/L): 49.18

Volume V (L): 5.80

t = 39.00 h

Biomass X (g/L): 88.39

Glucose S (g/L): 23.09

Lactic Acid P (g/L): 70.51

Maltose M (g/L): 47.99

Volume V (L): 5.70

t = 40.00 h

Biomass X (g/L): 89.32

Glucose S (g/L): 22.53

Lactic Acid P (g/L): 65.93

Maltose M (g/L): 48.98

Volume V (L): 5.86

t = 41.00 h

Biomass X (g/L): 87.68

Glucose S (g/L): 24.59

Lactic Acid P (g/L): 74.09

Maltose M (g/L): 49.90

Volume V (L): 5.75

t = 42.00 h

Biomass X (g/L): 83.53

Glucose S (g/L): 27.02

Lactic Acid P (g/L): 71.16

Maltose M (g/L): 44.78

Volume V (L): 5.89

t = 43.00 h

Biomass X (g/L): 92.34

Glucose S (g/L): 23.03

Lactic Acid P (g/L): 72.80

Maltose M (g/L): 47.11

Volume V (L): 5.91

t = 44.00 h

Biomass X (g/L): 92.12

Glucose S (g/L): 24.74

Lactic Acid P (g/L): 74.37

Maltose M (g/L): 46.21

Volume V (L): 5.83

t = 45.00 h

Biomass X (g/L): 89.61

Glucose S (g/L): 26.87

Lactic Acid P (g/L): 74.51

Maltose M (g/L): 46.91

Volume V (L): 5.88

t = 46.00 h

Biomass X (g/L): 90.64

Glucose S (g/L): 25.12

Lactic Acid P (g/L): 76.28

Maltose M (g/L): 45.77

Volume V (L): 5.87

t = 47.00 h

Biomass X (g/L): 91.91

Glucose S (g/L): 26.10

Lactic Acid P (g/L): 73.86

Maltose M (g/L): 43.91

Volume V (L): 5.93

t = 48.00 h

Biomass X (g/L): 86.57

Glucose S (g/L): 24.34

Lactic Acid P (g/L): 76.69

Maltose M (g/L): 44.07

Volume V (L): 5.95

t = 49.00 h

Biomass X (g/L): 89.33

Glucose S (g/L): 26.63

Lactic Acid P (g/L): 75.18

Maltose M (g/L): 42.93

Volume V (L): 6.02

t = 50.00 h

Biomass X (g/L): 92.82

Glucose S (g/L): 26.09

Lactic Acid P (g/L): 78.11

Maltose M (g/L): 45.16

Volume V (L): 6.06

t = 51.00 h

Biomass X (g/L): 87.51

Glucose S (g/L): 27.97

Lactic Acid P (g/L): 78.20

Maltose M (g/L): 43.21

Volume V (L): 6.08

t = 52.00 h

Biomass X (g/L): 91.56

Glucose S (g/L): 28.54
Lactic Acid P (g/L): 74.69
Maltose M (g/L): 45.47
Volume V (L): 6.01

t = 53.00 h

Biomass X (g/L): 87.26
Glucose S (g/L): 30.74
Lactic Acid P (g/L): 80.16
Maltose M (g/L): 42.24
Volume V (L): 6.05

t = 54.00 h

Biomass X (g/L): 92.00
Glucose S (g/L): 28.33
Lactic Acid P (g/L): 76.45
Maltose M (g/L): 40.90
Volume V (L): 6.00

t = 55.00 h

Biomass X (g/L): 90.67
Glucose S (g/L): 28.30
Lactic Acid P (g/L): 80.20
Maltose M (g/L): 39.35
Volume V (L): 6.06

t = 56.00 h

Biomass X (g/L): 86.52
Glucose S (g/L): 28.44
Lactic Acid P (g/L): 79.44
Maltose M (g/L): 38.53
Volume V (L): 6.16

t = 57.00 h

Biomass X (g/L): 93.71
Glucose S (g/L): 28.01
Lactic Acid P (g/L): 83.49
Maltose M (g/L): 38.46
Volume V (L): 6.21

t = 58.00 h

Biomass X (g/L): 88.71
Glucose S (g/L): 29.26
Lactic Acid P (g/L): 78.64
Maltose M (g/L): 40.72
Volume V (L): 6.10

t = 59.00 h

Biomass X (g/L): 88.45
Glucose S (g/L): 28.73
Lactic Acid P (g/L): 78.65
Maltose M (g/L): 40.69
Volume V (L): 6.15

t = 60.00 h

Biomass X (g/L): 91.00
Glucose S (g/L): 29.67
Lactic Acid P (g/L): 79.80
Maltose M (g/L): 37.34
Volume V (L): 6.20

t = 61.00 h

Biomass X (g/L): 93.39
Glucose S (g/L): 31.54
Lactic Acid P (g/L): 79.40
Maltose M (g/L): 40.52
Volume V (L): 6.23

t = 62.00 h

Biomass X (g/L): 87.66
Glucose S (g/L): 28.73
Lactic Acid P (g/L): 82.48
Maltose M (g/L): 38.19
Volume V (L): 6.20

t = 63.00 h

Biomass X (g/L): 90.37
Glucose S (g/L): 31.94
Lactic Acid P (g/L): 76.70
Maltose M (g/L): 38.74
Volume V (L): 6.28

t = 64.00 h

Biomass X (g/L): 91.22
Glucose S (g/L): 32.50
Lactic Acid P (g/L): 78.41
Maltose M (g/L): 40.50
Volume V (L): 6.26

t = 65.00 h

Biomass X (g/L): 91.79
Glucose S (g/L): 30.88
Lactic Acid P (g/L): 78.66
Maltose M (g/L): 39.14
Volume V (L): 6.30

t = 66.00 h

Biomass X (g/L): 90.33
Glucose S (g/L): 28.50
Lactic Acid P (g/L): 84.02
Maltose M (g/L): 37.14
Volume V (L): 6.34

t = 67.00 h

Biomass X (g/L): 88.79
Glucose S (g/L): 30.86
Lactic Acid P (g/L): 86.20
Maltose M (g/L): 35.93
Volume V (L): 6.34

t = 68.00 h

Biomass X (g/L): 87.46
Glucose S (g/L): 33.75
Lactic Acid P (g/L): 81.90
Maltose M (g/L): 33.81
Volume V (L): 6.34

t = 69.00 h

Biomass X (g/L): 89.70
Glucose S (g/L): 32.29
Lactic Acid P (g/L): 82.01
Maltose M (g/L): 35.55
Volume V (L): 6.38

t = 70.00 h

Biomass X (g/L): 86.54
Glucose S (g/L): 31.29
Lactic Acid P (g/L): 82.87
Maltose M (g/L): 35.97
Volume V (L): 6.40

t = 71.00 h

Biomass X (g/L): 81.58
Glucose S (g/L): 32.27
Lactic Acid P (g/L): 85.87
Maltose M (g/L): 34.56
Volume V (L): 6.40

t = 72.00 h

Biomass X (g/L): 90.17
Glucose S (g/L): 34.39
Lactic Acid P (g/L): 80.86
Maltose M (g/L): 35.49
Volume V (L): 6.43

t = 73.00 h

Biomass X (g/L): 91.24
Glucose S (g/L): 34.31
Lactic Acid P (g/L): 84.04
Maltose M (g/L): 35.50
Volume V (L): 6.53

t = 74.00 h

Biomass X (g/L): 82.57
Glucose S (g/L): 33.56
Lactic Acid P (g/L): 86.13
Maltose M (g/L): 34.78
Volume V (L): 6.51

t = 75.00 h

Biomass X (g/L): 85.93
Glucose S (g/L): 33.70
Lactic Acid P (g/L): 85.18
Maltose M (g/L): 33.90

Volume V (L): 6.49

Batch 6

Initial Conditions:

Batch: 6.00

X0 (g/L): 2.00

S0 (g/L): 33.00

M0 (g/L): 77.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.33

F_vol (L/h): 0.05

S_feed (g/L): 90.00

M_feed (g/L): 210.00

V_final (L): 8.75

Batch 6 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.67

Glucose S (g/L): 33.92

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 77.42

Volume V (L): 4.89

t = 1.00 h

Biomass X (g/L): 1.83

Glucose S (g/L): 35.18

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 77.25

Volume V (L): 5.10

t = 2.00 h

Biomass X (g/L): 6.26

Glucose S (g/L): 34.96

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 76.91

Volume V (L): 5.27

t = 3.00 h

Biomass X (g/L): 5.40

Glucose S (g/L): 36.84

Lactic Acid P (g/L): 0.02

Maltose M (g/L): 76.48

Volume V (L): 5.19

t = 4.00 h

Biomass X (g/L): 5.60

Glucose S (g/L): 36.32

Lactic Acid P (g/L): 1.78

Maltose M (g/L): 76.66

Volume V (L): 5.16

t = 5.00 h

Biomass X (g/L): 8.62

Glucose S (g/L): 36.23

Lactic Acid P (g/L): 6.04
Maltose M (g/L): 76.42
Volume V (L): 5.25

t = 6.00 h

Biomass X (g/L): 5.85
Glucose S (g/L): 36.68
Lactic Acid P (g/L): 6.17
Maltose M (g/L): 75.66
Volume V (L): 5.16

t = 7.00 h

Biomass X (g/L): 9.38
Glucose S (g/L): 37.30
Lactic Acid P (g/L): 7.82
Maltose M (g/L): 75.32
Volume V (L): 5.42

t = 8.00 h

Biomass X (g/L): 12.09
Glucose S (g/L): 36.35
Lactic Acid P (g/L): 11.39
Maltose M (g/L): 75.50
Volume V (L): 5.50

t = 9.00 h

Biomass X (g/L): 18.53
Glucose S (g/L): 34.55
Lactic Acid P (g/L): 11.55
Maltose M (g/L): 76.12
Volume V (L): 5.40

t = 10.00 h

Biomass X (g/L): 25.89
Glucose S (g/L): 32.94
Lactic Acid P (g/L): 15.35
Maltose M (g/L): 74.27
Volume V (L): 5.45

t = 11.00 h

Biomass X (g/L): 30.86

Glucose S (g/L): 30.72
Lactic Acid P (g/L): 17.02
Maltose M (g/L): 73.91
Volume V (L): 5.58

t = 12.00 h

Biomass X (g/L): 30.93
Glucose S (g/L): 27.58
Lactic Acid P (g/L): 20.73
Maltose M (g/L): 74.62
Volume V (L): 5.55

t = 13.00 h

Biomass X (g/L): 34.50
Glucose S (g/L): 26.20
Lactic Acid P (g/L): 21.73
Maltose M (g/L): 74.31
Volume V (L): 5.64

t = 14.00 h

Biomass X (g/L): 32.81
Glucose S (g/L): 24.28
Lactic Acid P (g/L): 20.42
Maltose M (g/L): 74.14
Volume V (L): 5.94

t = 15.00 h

Biomass X (g/L): 39.47
Glucose S (g/L): 22.14
Lactic Acid P (g/L): 28.35
Maltose M (g/L): 73.93
Volume V (L): 5.72

t = 16.00 h

Biomass X (g/L): 42.33
Glucose S (g/L): 20.54
Lactic Acid P (g/L): 34.47
Maltose M (g/L): 73.41
Volume V (L): 5.76

t = 17.00 h

Biomass X (g/L): 45.93
Glucose S (g/L): 18.40
Lactic Acid P (g/L): 35.83
Maltose M (g/L): 74.15
Volume V (L): 5.78

t = 18.00 h

Biomass X (g/L): 49.96
Glucose S (g/L): 16.56
Lactic Acid P (g/L): 38.23
Maltose M (g/L): 72.10
Volume V (L): 5.98

t = 19.00 h

Biomass X (g/L): 51.92
Glucose S (g/L): 15.82
Lactic Acid P (g/L): 40.97
Maltose M (g/L): 72.29
Volume V (L): 5.93

t = 20.00 h

Biomass X (g/L): 50.77
Glucose S (g/L): 13.79
Lactic Acid P (g/L): 40.37
Maltose M (g/L): 72.45
Volume V (L): 5.93

t = 21.00 h

Biomass X (g/L): 59.51
Glucose S (g/L): 12.16
Lactic Acid P (g/L): 42.44
Maltose M (g/L): 71.29
Volume V (L): 6.20

t = 22.00 h

Biomass X (g/L): 57.17
Glucose S (g/L): 11.98
Lactic Acid P (g/L): 42.30
Maltose M (g/L): 71.88
Volume V (L): 6.26

t = 23.00 h

Biomass X (g/L): 62.14
Glucose S (g/L): 10.89
Lactic Acid P (g/L): 44.53
Maltose M (g/L): 72.68
Volume V (L): 6.08

t = 24.00 h

Biomass X (g/L): 68.18
Glucose S (g/L): 10.47
Lactic Acid P (g/L): 48.07
Maltose M (g/L): 71.45
Volume V (L): 5.99

t = 25.00 h

Biomass X (g/L): 62.53
Glucose S (g/L): 10.54
Lactic Acid P (g/L): 50.74
Maltose M (g/L): 71.30
Volume V (L): 6.36

t = 26.00 h

Biomass X (g/L): 60.06
Glucose S (g/L): 9.75
Lactic Acid P (g/L): 48.98
Maltose M (g/L): 70.17
Volume V (L): 6.22

t = 27.00 h

Biomass X (g/L): 66.69
Glucose S (g/L): 9.98
Lactic Acid P (g/L): 49.47
Maltose M (g/L): 70.27
Volume V (L): 6.44

t = 28.00 h

Biomass X (g/L): 71.17
Glucose S (g/L): 8.60
Lactic Acid P (g/L): 51.59
Maltose M (g/L): 70.13
Volume V (L): 6.18

t = 29.00 h

Biomass X (g/L): 74.29
Glucose S (g/L): 10.30
Lactic Acid P (g/L): 53.99
Maltose M (g/L): 69.89
Volume V (L): 6.55

t = 30.00 h

Biomass X (g/L): 67.65
Glucose S (g/L): 10.16
Lactic Acid P (g/L): 50.40
Maltose M (g/L): 70.54
Volume V (L): 6.36

t = 31.00 h

Biomass X (g/L): 72.69
Glucose S (g/L): 9.58
Lactic Acid P (g/L): 49.44
Maltose M (g/L): 69.84
Volume V (L): 6.63

t = 32.00 h

Biomass X (g/L): 71.24
Glucose S (g/L): 10.92
Lactic Acid P (g/L): 54.26
Maltose M (g/L): 68.59
Volume V (L): 6.60

t = 33.00 h

Biomass X (g/L): 72.89
Glucose S (g/L): 11.32
Lactic Acid P (g/L): 53.62
Maltose M (g/L): 69.10
Volume V (L): 6.54

t = 34.00 h

Biomass X (g/L): 80.20
Glucose S (g/L): 9.45
Lactic Acid P (g/L): 56.04
Maltose M (g/L): 68.97

Volume V (L): 6.65

t = 35.00 h

Biomass X (g/L): 70.96

Glucose S (g/L): 10.38

Lactic Acid P (g/L): 57.64

Maltose M (g/L): 69.06

Volume V (L): 6.83

t = 36.00 h

Biomass X (g/L): 75.80

Glucose S (g/L): 10.64

Lactic Acid P (g/L): 57.71

Maltose M (g/L): 69.04

Volume V (L): 6.81

t = 37.00 h

Biomass X (g/L): 76.90

Glucose S (g/L): 10.80

Lactic Acid P (g/L): 62.72

Maltose M (g/L): 67.25

Volume V (L): 6.92

t = 38.00 h

Biomass X (g/L): 78.94

Glucose S (g/L): 12.56

Lactic Acid P (g/L): 60.89

Maltose M (g/L): 67.09

Volume V (L): 7.12

t = 39.00 h

Biomass X (g/L): 77.09

Glucose S (g/L): 12.07

Lactic Acid P (g/L): 61.51

Maltose M (g/L): 66.55

Volume V (L): 6.77

t = 40.00 h

Biomass X (g/L): 78.07

Glucose S (g/L): 12.37

Lactic Acid P (g/L): 62.88

Maltose M (g/L): 66.86

Volume V (L): 7.18

t = 41.00 h

Biomass X (g/L): 76.84

Glucose S (g/L): 11.33

Lactic Acid P (g/L): 59.56

Maltose M (g/L): 66.99

Volume V (L): 7.06

t = 42.00 h

Biomass X (g/L): 78.83

Glucose S (g/L): 13.88

Lactic Acid P (g/L): 65.05

Maltose M (g/L): 67.01

Volume V (L): 7.00

t = 43.00 h

Biomass X (g/L): 77.75

Glucose S (g/L): 12.76

Lactic Acid P (g/L): 63.13

Maltose M (g/L): 66.12

Volume V (L): 7.00

t = 44.00 h

Biomass X (g/L): 80.41

Glucose S (g/L): 14.25

Lactic Acid P (g/L): 65.37

Maltose M (g/L): 65.70

Volume V (L): 7.18

t = 45.00 h

Biomass X (g/L): 78.37

Glucose S (g/L): 14.86

Lactic Acid P (g/L): 62.01

Maltose M (g/L): 65.69

Volume V (L): 7.35

t = 46.00 h

Biomass X (g/L): 80.46

Glucose S (g/L): 16.20

Lactic Acid P (g/L): 61.45

Maltose M (g/L): 64.84

Volume V (L): 7.29

t = 47.00 h

Biomass X (g/L): 79.76

Glucose S (g/L): 14.48

Lactic Acid P (g/L): 67.28

Maltose M (g/L): 66.26

Volume V (L): 7.52

t = 48.00 h

Biomass X (g/L): 82.78

Glucose S (g/L): 13.56

Lactic Acid P (g/L): 66.78

Maltose M (g/L): 63.83

Volume V (L): 7.24

t = 49.00 h

Biomass X (g/L): 80.19

Glucose S (g/L): 16.66

Lactic Acid P (g/L): 64.39

Maltose M (g/L): 64.60

Volume V (L): 7.45

t = 50.00 h

Biomass X (g/L): 76.29

Glucose S (g/L): 18.48

Lactic Acid P (g/L): 64.92

Maltose M (g/L): 65.39

Volume V (L): 7.36

t = 51.00 h

Biomass X (g/L): 83.74

Glucose S (g/L): 18.52

Lactic Acid P (g/L): 69.84

Maltose M (g/L): 63.94

Volume V (L): 7.59

t = 52.00 h

Biomass X (g/L): 78.72

Glucose S (g/L): 18.19
Lactic Acid P (g/L): 65.07
Maltose M (g/L): 63.56
Volume V (L): 7.70

t = 53.00 h

Biomass X (g/L): 79.48
Glucose S (g/L): 18.13
Lactic Acid P (g/L): 71.84
Maltose M (g/L): 63.18
Volume V (L): 7.60

t = 54.00 h

Biomass X (g/L): 79.75
Glucose S (g/L): 19.72
Lactic Acid P (g/L): 66.54
Maltose M (g/L): 63.01
Volume V (L): 7.41

t = 55.00 h

Biomass X (g/L): 84.58
Glucose S (g/L): 18.79
Lactic Acid P (g/L): 70.38
Maltose M (g/L): 63.65
Volume V (L): 7.77

t = 56.00 h

Biomass X (g/L): 76.29
Glucose S (g/L): 20.51
Lactic Acid P (g/L): 68.59
Maltose M (g/L): 61.29
Volume V (L): 7.85

t = 57.00 h

Biomass X (g/L): 88.99
Glucose S (g/L): 21.08
Lactic Acid P (g/L): 71.05
Maltose M (g/L): 63.20
Volume V (L): 7.90

t = 58.00 h

Biomass X (g/L): 84.08
Glucose S (g/L): 22.08
Lactic Acid P (g/L): 68.83
Maltose M (g/L): 61.21
Volume V (L): 8.04

t = 59.00 h

Biomass X (g/L): 82.15
Glucose S (g/L): 24.01
Lactic Acid P (g/L): 70.03
Maltose M (g/L): 61.61
Volume V (L): 7.83

t = 60.00 h

Biomass X (g/L): 81.53
Glucose S (g/L): 22.63
Lactic Acid P (g/L): 70.82
Maltose M (g/L): 61.25
Volume V (L): 8.08

t = 61.00 h

Biomass X (g/L): 82.38
Glucose S (g/L): 22.45
Lactic Acid P (g/L): 68.81
Maltose M (g/L): 61.58
Volume V (L): 8.18

t = 62.00 h

Biomass X (g/L): 82.97
Glucose S (g/L): 23.41
Lactic Acid P (g/L): 73.03
Maltose M (g/L): 60.04
Volume V (L): 7.90

t = 63.00 h

Biomass X (g/L): 83.28
Glucose S (g/L): 22.54
Lactic Acid P (g/L): 70.53
Maltose M (g/L): 60.97
Volume V (L): 8.19

t = 64.00 h

Biomass X (g/L): 83.06
Glucose S (g/L): 23.99
Lactic Acid P (g/L): 71.46
Maltose M (g/L): 61.62
Volume V (L): 8.14

t = 65.00 h

Biomass X (g/L): 81.94
Glucose S (g/L): 24.78
Lactic Acid P (g/L): 70.04
Maltose M (g/L): 59.39
Volume V (L): 8.24

t = 66.00 h

Biomass X (g/L): 81.24
Glucose S (g/L): 25.26
Lactic Acid P (g/L): 66.80
Maltose M (g/L): 60.03
Volume V (L): 8.34

t = 67.00 h

Biomass X (g/L): 81.32
Glucose S (g/L): 26.38
Lactic Acid P (g/L): 72.39
Maltose M (g/L): 59.28
Volume V (L): 8.30

t = 68.00 h

Biomass X (g/L): 81.18
Glucose S (g/L): 26.15
Lactic Acid P (g/L): 78.50
Maltose M (g/L): 59.65
Volume V (L): 8.32

t = 69.00 h

Biomass X (g/L): 84.31
Glucose S (g/L): 28.31
Lactic Acid P (g/L): 75.10
Maltose M (g/L): 59.16
Volume V (L): 8.45

t = 70.00 h

Biomass X (g/L): 82.97
Glucose S (g/L): 27.97
Lactic Acid P (g/L): 73.55
Maltose M (g/L): 58.52
Volume V (L): 8.38

t = 71.00 h

Biomass X (g/L): 74.45
Glucose S (g/L): 28.53
Lactic Acid P (g/L): 73.25
Maltose M (g/L): 58.92
Volume V (L): 8.43

t = 72.00 h

Biomass X (g/L): 82.99
Glucose S (g/L): 28.84
Lactic Acid P (g/L): 74.13
Maltose M (g/L): 59.43
Volume V (L): 8.68

t = 73.00 h

Biomass X (g/L): 81.81
Glucose S (g/L): 29.80
Lactic Acid P (g/L): 75.54
Maltose M (g/L): 58.02
Volume V (L): 8.52

t = 74.00 h

Biomass X (g/L): 81.22
Glucose S (g/L): 30.92
Lactic Acid P (g/L): 71.89
Maltose M (g/L): 58.30
Volume V (L): 8.77

t = 75.00 h

Biomass X (g/L): 80.25
Glucose S (g/L): 30.11
Lactic Acid P (g/L): 77.26
Maltose M (g/L): 56.88

Volume V (L): 8.80

Batch 7

Initial Conditions:

Batch: 7.00

X0 (g/L): 3.50

S0 (g/L): 112.00

M0 (g/L): 48.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.43

F_vol (L/h): 0.04

S_feed (g/L): 210.00

M_feed (g/L): 90.00

V_final (L): 8.38

Batch 7 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 111.49
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 47.10
Volume V (L): 5.11

t = 1.00 h

Biomass X (g/L): 3.05
Glucose S (g/L): 112.71
Lactic Acid P (g/L): 0.52
Maltose M (g/L): 46.57
Volume V (L): 4.98

t = 2.00 h

Biomass X (g/L): 9.45
Glucose S (g/L): 110.15
Lactic Acid P (g/L): 2.68
Maltose M (g/L): 47.24
Volume V (L): 4.96

t = 3.00 h

Biomass X (g/L): 6.06
Glucose S (g/L): 112.75
Lactic Acid P (g/L): 2.83
Maltose M (g/L): 46.76
Volume V (L): 5.11

t = 4.00 h

Biomass X (g/L): 10.02
Glucose S (g/L): 109.80
Lactic Acid P (g/L): 5.33
Maltose M (g/L): 46.24
Volume V (L): 5.10

t = 5.00 h

Biomass X (g/L): 15.57
Glucose S (g/L): 107.80

Lactic Acid P (g/L): 9.94
Maltose M (g/L): 45.20
Volume V (L): 5.22

t = 6.00 h

Biomass X (g/L): 20.13
Glucose S (g/L): 105.43
Lactic Acid P (g/L): 9.68
Maltose M (g/L): 44.80
Volume V (L): 5.40

t = 7.00 h

Biomass X (g/L): 21.71
Glucose S (g/L): 106.42
Lactic Acid P (g/L): 9.99
Maltose M (g/L): 44.62
Volume V (L): 5.27

t = 8.00 h

Biomass X (g/L): 25.39
Glucose S (g/L): 101.00
Lactic Acid P (g/L): 21.29
Maltose M (g/L): 43.99
Volume V (L): 5.36

t = 9.00 h

Biomass X (g/L): 35.25
Glucose S (g/L): 100.70
Lactic Acid P (g/L): 23.92
Maltose M (g/L): 41.99
Volume V (L): 5.41

t = 10.00 h

Biomass X (g/L): 40.85
Glucose S (g/L): 94.48
Lactic Acid P (g/L): 21.13
Maltose M (g/L): 42.86
Volume V (L): 5.54

t = 11.00 h

Biomass X (g/L): 43.62

Glucose S (g/L): 89.52
Lactic Acid P (g/L): 29.34
Maltose M (g/L): 42.00
Volume V (L): 5.27

t = 12.00 h

Biomass X (g/L): 45.86
Glucose S (g/L): 87.14
Lactic Acid P (g/L): 29.90
Maltose M (g/L): 42.25
Volume V (L): 5.48

t = 13.00 h

Biomass X (g/L): 47.83
Glucose S (g/L): 87.38
Lactic Acid P (g/L): 34.94
Maltose M (g/L): 43.70
Volume V (L): 5.61

t = 14.00 h

Biomass X (g/L): 51.70
Glucose S (g/L): 82.73
Lactic Acid P (g/L): 38.03
Maltose M (g/L): 41.54
Volume V (L): 5.75

t = 15.00 h

Biomass X (g/L): 58.73
Glucose S (g/L): 79.72
Lactic Acid P (g/L): 39.58
Maltose M (g/L): 40.68
Volume V (L): 5.63

t = 16.00 h

Biomass X (g/L): 61.74
Glucose S (g/L): 78.49
Lactic Acid P (g/L): 40.87
Maltose M (g/L): 41.10
Volume V (L): 5.53

t = 17.00 h

Biomass X (g/L): 67.20
Glucose S (g/L): 77.02
Lactic Acid P (g/L): 43.48
Maltose M (g/L): 40.41
Volume V (L): 5.83

t = 18.00 h

Biomass X (g/L): 67.98
Glucose S (g/L): 76.18
Lactic Acid P (g/L): 45.14
Maltose M (g/L): 39.99
Volume V (L): 5.75

t = 19.00 h

Biomass X (g/L): 64.93
Glucose S (g/L): 75.16
Lactic Acid P (g/L): 47.35
Maltose M (g/L): 39.73
Volume V (L): 5.73

t = 20.00 h

Biomass X (g/L): 73.57
Glucose S (g/L): 71.41
Lactic Acid P (g/L): 51.01
Maltose M (g/L): 40.33
Volume V (L): 5.84

t = 21.00 h

Biomass X (g/L): 70.33
Glucose S (g/L): 69.00
Lactic Acid P (g/L): 51.18
Maltose M (g/L): 37.84
Volume V (L): 5.93

t = 22.00 h

Biomass X (g/L): 73.63
Glucose S (g/L): 69.34
Lactic Acid P (g/L): 56.06
Maltose M (g/L): 38.55
Volume V (L): 5.93

t = 23.00 h

Biomass X (g/L): 76.80
Glucose S (g/L): 69.50
Lactic Acid P (g/L): 59.41
Maltose M (g/L): 37.27
Volume V (L): 5.81

t = 24.00 h

Biomass X (g/L): 82.63
Glucose S (g/L): 67.18
Lactic Acid P (g/L): 55.84
Maltose M (g/L): 37.91
Volume V (L): 6.17

t = 25.00 h

Biomass X (g/L): 86.71
Glucose S (g/L): 66.67
Lactic Acid P (g/L): 59.02
Maltose M (g/L): 36.59
Volume V (L): 6.25

t = 26.00 h

Biomass X (g/L): 81.50
Glucose S (g/L): 68.05
Lactic Acid P (g/L): 60.79
Maltose M (g/L): 36.59
Volume V (L): 6.24

t = 27.00 h

Biomass X (g/L): 80.51
Glucose S (g/L): 67.18
Lactic Acid P (g/L): 61.79
Maltose M (g/L): 36.35
Volume V (L): 6.10

t = 28.00 h

Biomass X (g/L): 83.24
Glucose S (g/L): 67.05
Lactic Acid P (g/L): 61.85
Maltose M (g/L): 36.88
Volume V (L): 6.24

t = 29.00 h

Biomass X (g/L): 84.94
Glucose S (g/L): 65.80
Lactic Acid P (g/L): 64.35
Maltose M (g/L): 36.52
Volume V (L): 6.30

t = 30.00 h

Biomass X (g/L): 82.60
Glucose S (g/L): 66.42
Lactic Acid P (g/L): 64.61
Maltose M (g/L): 34.77
Volume V (L): 6.41

t = 31.00 h

Biomass X (g/L): 88.61
Glucose S (g/L): 65.42
Lactic Acid P (g/L): 61.34
Maltose M (g/L): 35.34
Volume V (L): 6.46

t = 32.00 h

Biomass X (g/L): 85.22
Glucose S (g/L): 67.93
Lactic Acid P (g/L): 67.11
Maltose M (g/L): 34.53
Volume V (L): 6.37

t = 33.00 h

Biomass X (g/L): 86.41
Glucose S (g/L): 68.18
Lactic Acid P (g/L): 67.87
Maltose M (g/L): 33.87
Volume V (L): 6.49

t = 34.00 h

Biomass X (g/L): 87.29
Glucose S (g/L): 64.31
Lactic Acid P (g/L): 64.64
Maltose M (g/L): 34.27

Volume V (L): 6.58

t = 35.00 h

Biomass X (g/L): 87.07

Glucose S (g/L): 65.71

Lactic Acid P (g/L): 67.18

Maltose M (g/L): 33.93

Volume V (L): 6.72

t = 36.00 h

Biomass X (g/L): 88.48

Glucose S (g/L): 65.09

Lactic Acid P (g/L): 69.31

Maltose M (g/L): 35.48

Volume V (L): 6.69

t = 37.00 h

Biomass X (g/L): 85.10

Glucose S (g/L): 66.45

Lactic Acid P (g/L): 67.77

Maltose M (g/L): 33.87

Volume V (L): 6.74

t = 38.00 h

Biomass X (g/L): 85.51

Glucose S (g/L): 67.46

Lactic Acid P (g/L): 71.36

Maltose M (g/L): 32.40

Volume V (L): 6.72

t = 39.00 h

Biomass X (g/L): 84.37

Glucose S (g/L): 65.82

Lactic Acid P (g/L): 74.94

Maltose M (g/L): 32.67

Volume V (L): 6.90

t = 40.00 h

Biomass X (g/L): 88.46

Glucose S (g/L): 66.81

Lactic Acid P (g/L): 67.96

Maltose M (g/L): 33.47

Volume V (L): 6.84

t = 41.00 h

Biomass X (g/L): 89.49

Glucose S (g/L): 66.99

Lactic Acid P (g/L): 67.02

Maltose M (g/L): 33.93

Volume V (L): 6.76

t = 42.00 h

Biomass X (g/L): 88.81

Glucose S (g/L): 68.96

Lactic Acid P (g/L): 71.46

Maltose M (g/L): 31.22

Volume V (L): 7.03

t = 43.00 h

Biomass X (g/L): 92.00

Glucose S (g/L): 69.09

Lactic Acid P (g/L): 72.88

Maltose M (g/L): 30.78

Volume V (L): 7.01

t = 44.00 h

Biomass X (g/L): 87.82

Glucose S (g/L): 68.33

Lactic Acid P (g/L): 71.03

Maltose M (g/L): 33.04

Volume V (L): 6.94

t = 45.00 h

Biomass X (g/L): 89.59

Glucose S (g/L): 69.91

Lactic Acid P (g/L): 75.42

Maltose M (g/L): 31.81

Volume V (L): 6.98

t = 46.00 h

Biomass X (g/L): 85.17

Glucose S (g/L): 69.27

Lactic Acid P (g/L): 69.17
Maltose M (g/L): 30.73
Volume V (L): 7.10

t = 47.00 h

Biomass X (g/L): 86.97
Glucose S (g/L): 70.01
Lactic Acid P (g/L): 74.35
Maltose M (g/L): 31.43
Volume V (L): 7.16

t = 48.00 h

Biomass X (g/L): 89.72
Glucose S (g/L): 73.06
Lactic Acid P (g/L): 71.61
Maltose M (g/L): 29.09
Volume V (L): 7.18

t = 49.00 h

Biomass X (g/L): 90.09
Glucose S (g/L): 72.98
Lactic Acid P (g/L): 77.46
Maltose M (g/L): 30.63
Volume V (L): 7.24

t = 50.00 h

Biomass X (g/L): 87.16
Glucose S (g/L): 69.73
Lactic Acid P (g/L): 78.42
Maltose M (g/L): 30.62
Volume V (L): 7.42

t = 51.00 h

Biomass X (g/L): 87.68
Glucose S (g/L): 73.06
Lactic Acid P (g/L): 72.19
Maltose M (g/L): 30.80
Volume V (L): 7.40

t = 52.00 h

Biomass X (g/L): 83.96

Glucose S (g/L): 73.43
Lactic Acid P (g/L): 73.00
Maltose M (g/L): 29.29
Volume V (L): 7.36

t = 53.00 h

Biomass X (g/L): 86.83
Glucose S (g/L): 71.13
Lactic Acid P (g/L): 76.63
Maltose M (g/L): 29.81
Volume V (L): 7.48

t = 54.00 h

Biomass X (g/L): 89.91
Glucose S (g/L): 74.25
Lactic Acid P (g/L): 73.94
Maltose M (g/L): 28.93
Volume V (L): 7.33

t = 55.00 h

Biomass X (g/L): 89.02
Glucose S (g/L): 75.64
Lactic Acid P (g/L): 75.02
Maltose M (g/L): 29.51
Volume V (L): 7.52

t = 56.00 h

Biomass X (g/L): 88.72
Glucose S (g/L): 72.39
Lactic Acid P (g/L): 75.44
Maltose M (g/L): 28.54
Volume V (L): 7.70

t = 57.00 h

Biomass X (g/L): 87.47
Glucose S (g/L): 73.09
Lactic Acid P (g/L): 71.08
Maltose M (g/L): 28.66
Volume V (L): 7.54

t = 58.00 h

Biomass X (g/L): 87.24
Glucose S (g/L): 76.17
Lactic Acid P (g/L): 75.84
Maltose M (g/L): 28.07
Volume V (L): 7.52

t = 59.00 h

Biomass X (g/L): 85.42
Glucose S (g/L): 75.55
Lactic Acid P (g/L): 74.47
Maltose M (g/L): 28.98
Volume V (L): 7.59

t = 60.00 h

Biomass X (g/L): 88.73
Glucose S (g/L): 76.36
Lactic Acid P (g/L): 75.16
Maltose M (g/L): 28.92
Volume V (L): 7.59

t = 61.00 h

Biomass X (g/L): 87.88
Glucose S (g/L): 76.48
Lactic Acid P (g/L): 79.97
Maltose M (g/L): 28.00
Volume V (L): 7.67

t = 62.00 h

Biomass X (g/L): 87.91
Glucose S (g/L): 76.97
Lactic Acid P (g/L): 73.78
Maltose M (g/L): 28.00
Volume V (L): 7.75

t = 63.00 h

Biomass X (g/L): 85.18
Glucose S (g/L): 75.02
Lactic Acid P (g/L): 78.89
Maltose M (g/L): 26.82
Volume V (L): 7.69

t = 64.00 h

Biomass X (g/L): 80.89
Glucose S (g/L): 78.71
Lactic Acid P (g/L): 76.50
Maltose M (g/L): 27.41
Volume V (L): 7.88

t = 65.00 h

Biomass X (g/L): 88.96
Glucose S (g/L): 79.29
Lactic Acid P (g/L): 76.69
Maltose M (g/L): 27.40
Volume V (L): 8.02

t = 66.00 h

Biomass X (g/L): 84.80
Glucose S (g/L): 77.78
Lactic Acid P (g/L): 79.42
Maltose M (g/L): 29.00
Volume V (L): 7.88

t = 67.00 h

Biomass X (g/L): 83.65
Glucose S (g/L): 78.45
Lactic Acid P (g/L): 78.53
Maltose M (g/L): 26.34
Volume V (L): 8.17

t = 68.00 h

Biomass X (g/L): 87.41
Glucose S (g/L): 76.82
Lactic Acid P (g/L): 79.26
Maltose M (g/L): 27.89
Volume V (L): 8.11

t = 69.00 h

Biomass X (g/L): 84.48
Glucose S (g/L): 80.05
Lactic Acid P (g/L): 80.38
Maltose M (g/L): 27.40
Volume V (L): 8.21

t = 70.00 h

Biomass X (g/L): 87.97
Glucose S (g/L): 80.54
Lactic Acid P (g/L): 77.46
Maltose M (g/L): 26.76
Volume V (L): 8.08

t = 71.00 h

Biomass X (g/L): 86.66
Glucose S (g/L): 80.07
Lactic Acid P (g/L): 76.96
Maltose M (g/L): 27.58
Volume V (L): 8.28

t = 72.00 h

Biomass X (g/L): 86.63
Glucose S (g/L): 82.12
Lactic Acid P (g/L): 76.00
Maltose M (g/L): 26.49
Volume V (L): 8.13

t = 73.00 h

Biomass X (g/L): 81.24
Glucose S (g/L): 80.51
Lactic Acid P (g/L): 76.77
Maltose M (g/L): 26.34
Volume V (L): 8.38

t = 74.00 h

Biomass X (g/L): 87.57
Glucose S (g/L): 82.16
Lactic Acid P (g/L): 85.88
Maltose M (g/L): 25.89
Volume V (L): 8.35

t = 75.00 h

Biomass X (g/L): 84.96
Glucose S (g/L): 80.38
Lactic Acid P (g/L): 81.98
Maltose M (g/L): 26.58

Volume V (L): 8.43

Batch 8

Initial Conditions:

Batch: 8.00

X0 (g/L): 2.00

S0 (g/L): 68.75

M0 (g/L): 56.25

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.82

F_vol (L/h): 0.06

S_feed (g/L): 165.00

M_feed (g/L): 135.00

V_final (L): 9.50

Batch 8 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 68.42

Lactic Acid P (g/L): 3.97

Maltose M (g/L): 56.07

Volume V (L): 5.17

t = 1.00 h

Biomass X (g/L): 0.48

Glucose S (g/L): 70.61

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 56.28

Volume V (L): 5.03

t = 2.00 h

Biomass X (g/L): 5.08

Glucose S (g/L): 73.06

Lactic Acid P (g/L): 5.37

Maltose M (g/L): 56.01

Volume V (L): 5.17

t = 3.00 h

Biomass X (g/L): 2.46

Glucose S (g/L): 73.30

Lactic Acid P (g/L): 3.93

Maltose M (g/L): 54.78

Volume V (L): 5.10

t = 4.00 h

Biomass X (g/L): 7.48

Glucose S (g/L): 73.42

Lactic Acid P (g/L): 4.29

Maltose M (g/L): 55.97

Volume V (L): 4.99

t = 5.00 h

Biomass X (g/L): 4.89

Glucose S (g/L): 73.58

Lactic Acid P (g/L): 5.45
Maltose M (g/L): 55.54
Volume V (L): 5.34

t = 6.00 h

Biomass X (g/L): 12.75
Glucose S (g/L): 72.79
Lactic Acid P (g/L): 7.25
Maltose M (g/L): 55.50
Volume V (L): 5.10

t = 7.00 h

Biomass X (g/L): 10.47
Glucose S (g/L): 72.43
Lactic Acid P (g/L): 4.03
Maltose M (g/L): 56.92
Volume V (L): 5.41

t = 8.00 h

Biomass X (g/L): 18.20
Glucose S (g/L): 67.96
Lactic Acid P (g/L): 10.22
Maltose M (g/L): 53.91
Volume V (L): 5.30

t = 9.00 h

Biomass X (g/L): 22.11
Glucose S (g/L): 68.35
Lactic Acid P (g/L): 15.52
Maltose M (g/L): 55.51
Volume V (L): 5.81

t = 10.00 h

Biomass X (g/L): 22.90
Glucose S (g/L): 66.67
Lactic Acid P (g/L): 16.45
Maltose M (g/L): 54.37
Volume V (L): 5.72

t = 11.00 h

Biomass X (g/L): 30.32

Glucose S (g/L): 62.81
Lactic Acid P (g/L): 16.50
Maltose M (g/L): 54.44
Volume V (L): 5.67

t = 12.00 h

Biomass X (g/L): 33.87
Glucose S (g/L): 61.73
Lactic Acid P (g/L): 22.81
Maltose M (g/L): 54.08
Volume V (L): 5.89

t = 13.00 h

Biomass X (g/L): 36.21
Glucose S (g/L): 58.51
Lactic Acid P (g/L): 30.03
Maltose M (g/L): 53.70
Volume V (L): 5.67

t = 14.00 h

Biomass X (g/L): 39.23
Glucose S (g/L): 57.89
Lactic Acid P (g/L): 32.13
Maltose M (g/L): 53.42
Volume V (L): 5.91

t = 15.00 h

Biomass X (g/L): 40.85
Glucose S (g/L): 55.29
Lactic Acid P (g/L): 32.15
Maltose M (g/L): 53.64
Volume V (L): 5.93

t = 16.00 h

Biomass X (g/L): 50.30
Glucose S (g/L): 51.03
Lactic Acid P (g/L): 37.32
Maltose M (g/L): 52.56
Volume V (L): 6.00

t = 17.00 h

Biomass X (g/L): 49.03
Glucose S (g/L): 49.33
Lactic Acid P (g/L): 36.99
Maltose M (g/L): 53.41
Volume V (L): 6.19

t = 18.00 h

Biomass X (g/L): 56.31
Glucose S (g/L): 47.63
Lactic Acid P (g/L): 36.43
Maltose M (g/L): 52.65
Volume V (L): 6.18

t = 19.00 h

Biomass X (g/L): 58.33
Glucose S (g/L): 45.69
Lactic Acid P (g/L): 40.52
Maltose M (g/L): 52.10
Volume V (L): 6.18

t = 20.00 h

Biomass X (g/L): 63.30
Glucose S (g/L): 44.99
Lactic Acid P (g/L): 48.33
Maltose M (g/L): 52.20
Volume V (L): 5.98

t = 21.00 h

Biomass X (g/L): 63.27
Glucose S (g/L): 42.90
Lactic Acid P (g/L): 46.11
Maltose M (g/L): 52.78
Volume V (L): 6.13

t = 22.00 h

Biomass X (g/L): 65.36
Glucose S (g/L): 41.58
Lactic Acid P (g/L): 48.00
Maltose M (g/L): 51.72
Volume V (L): 6.30

t = 23.00 h

Biomass X (g/L): 69.94
Glucose S (g/L): 42.54
Lactic Acid P (g/L): 49.54
Maltose M (g/L): 51.92
Volume V (L): 6.39

t = 24.00 h

Biomass X (g/L): 70.73
Glucose S (g/L): 41.91
Lactic Acid P (g/L): 56.12
Maltose M (g/L): 51.18
Volume V (L): 6.29

t = 25.00 h

Biomass X (g/L): 72.27
Glucose S (g/L): 39.99
Lactic Acid P (g/L): 56.92
Maltose M (g/L): 51.66
Volume V (L): 6.83

t = 26.00 h

Biomass X (g/L): 74.09
Glucose S (g/L): 41.31
Lactic Acid P (g/L): 59.59
Maltose M (g/L): 51.40
Volume V (L): 6.76

t = 27.00 h

Biomass X (g/L): 70.43
Glucose S (g/L): 39.53
Lactic Acid P (g/L): 55.40
Maltose M (g/L): 51.72
Volume V (L): 6.70

t = 28.00 h

Biomass X (g/L): 77.26
Glucose S (g/L): 40.25
Lactic Acid P (g/L): 57.58
Maltose M (g/L): 50.54
Volume V (L): 6.69

t = 29.00 h

Biomass X (g/L): 77.96
Glucose S (g/L): 40.74
Lactic Acid P (g/L): 54.51
Maltose M (g/L): 50.71
Volume V (L): 6.71

t = 30.00 h

Biomass X (g/L): 74.88
Glucose S (g/L): 39.89
Lactic Acid P (g/L): 62.33
Maltose M (g/L): 50.22
Volume V (L): 6.67

t = 31.00 h

Biomass X (g/L): 77.65
Glucose S (g/L): 40.45
Lactic Acid P (g/L): 57.23
Maltose M (g/L): 50.56
Volume V (L): 6.87

t = 32.00 h

Biomass X (g/L): 78.92
Glucose S (g/L): 39.62
Lactic Acid P (g/L): 61.26
Maltose M (g/L): 49.38
Volume V (L): 6.95

t = 33.00 h

Biomass X (g/L): 79.38
Glucose S (g/L): 40.13
Lactic Acid P (g/L): 59.07
Maltose M (g/L): 49.84
Volume V (L): 6.81

t = 34.00 h

Biomass X (g/L): 81.16
Glucose S (g/L): 40.35
Lactic Acid P (g/L): 61.00
Maltose M (g/L): 49.18

Volume V (L): 7.09

t = 35.00 h

Biomass X (g/L): 84.43

Glucose S (g/L): 40.10

Lactic Acid P (g/L): 61.16

Maltose M (g/L): 49.01

Volume V (L): 7.08

t = 36.00 h

Biomass X (g/L): 84.07

Glucose S (g/L): 40.99

Lactic Acid P (g/L): 62.36

Maltose M (g/L): 48.86

Volume V (L): 6.90

t = 37.00 h

Biomass X (g/L): 78.60

Glucose S (g/L): 39.97

Lactic Acid P (g/L): 63.16

Maltose M (g/L): 48.71

Volume V (L): 7.10

t = 38.00 h

Biomass X (g/L): 83.78

Glucose S (g/L): 41.41

Lactic Acid P (g/L): 63.75

Maltose M (g/L): 48.31

Volume V (L): 7.22

t = 39.00 h

Biomass X (g/L): 85.88

Glucose S (g/L): 42.03

Lactic Acid P (g/L): 67.48

Maltose M (g/L): 48.24

Volume V (L): 7.59

t = 40.00 h

Biomass X (g/L): 83.86

Glucose S (g/L): 42.50

Lactic Acid P (g/L): 63.61

Maltose M (g/L): 48.66

Volume V (L): 7.48

t = 41.00 h

Biomass X (g/L): 85.88

Glucose S (g/L): 42.97

Lactic Acid P (g/L): 67.15

Maltose M (g/L): 48.15

Volume V (L): 7.28

t = 42.00 h

Biomass X (g/L): 88.40

Glucose S (g/L): 43.63

Lactic Acid P (g/L): 69.92

Maltose M (g/L): 47.87

Volume V (L): 7.59

t = 43.00 h

Biomass X (g/L): 86.16

Glucose S (g/L): 44.66

Lactic Acid P (g/L): 66.44

Maltose M (g/L): 47.03

Volume V (L): 7.37

t = 44.00 h

Biomass X (g/L): 84.42

Glucose S (g/L): 45.56

Lactic Acid P (g/L): 66.43

Maltose M (g/L): 46.60

Volume V (L): 7.79

t = 45.00 h

Biomass X (g/L): 86.54

Glucose S (g/L): 44.57

Lactic Acid P (g/L): 67.97

Maltose M (g/L): 47.25

Volume V (L): 7.64

t = 46.00 h

Biomass X (g/L): 84.45

Glucose S (g/L): 44.55

Lactic Acid P (g/L): 71.60
Maltose M (g/L): 46.99
Volume V (L): 7.75

t = 47.00 h

Biomass X (g/L): 84.25
Glucose S (g/L): 45.26
Lactic Acid P (g/L): 69.74
Maltose M (g/L): 47.05
Volume V (L): 8.00

t = 48.00 h

Biomass X (g/L): 83.40
Glucose S (g/L): 46.36
Lactic Acid P (g/L): 68.46
Maltose M (g/L): 45.85
Volume V (L): 7.71

t = 49.00 h

Biomass X (g/L): 88.20
Glucose S (g/L): 47.06
Lactic Acid P (g/L): 71.14
Maltose M (g/L): 45.81
Volume V (L): 7.75

t = 50.00 h

Biomass X (g/L): 82.36
Glucose S (g/L): 47.21
Lactic Acid P (g/L): 67.86
Maltose M (g/L): 44.92
Volume V (L): 7.92

t = 51.00 h

Biomass X (g/L): 88.19
Glucose S (g/L): 47.26
Lactic Acid P (g/L): 72.27
Maltose M (g/L): 45.29
Volume V (L): 8.20

t = 52.00 h

Biomass X (g/L): 85.65

Glucose S (g/L): 48.68
Lactic Acid P (g/L): 71.96
Maltose M (g/L): 45.73
Volume V (L): 7.91

t = 53.00 h

Biomass X (g/L): 80.95
Glucose S (g/L): 49.90
Lactic Acid P (g/L): 70.64
Maltose M (g/L): 44.83
Volume V (L): 7.80

t = 54.00 h

Biomass X (g/L): 85.30
Glucose S (g/L): 50.89
Lactic Acid P (g/L): 69.72
Maltose M (g/L): 45.13
Volume V (L): 8.18

t = 55.00 h

Biomass X (g/L): 79.17
Glucose S (g/L): 50.33
Lactic Acid P (g/L): 69.90
Maltose M (g/L): 44.53
Volume V (L): 8.37

t = 56.00 h

Biomass X (g/L): 85.87
Glucose S (g/L): 51.07
Lactic Acid P (g/L): 73.42
Maltose M (g/L): 44.56
Volume V (L): 8.52

t = 57.00 h

Biomass X (g/L): 86.46
Glucose S (g/L): 51.15
Lactic Acid P (g/L): 75.26
Maltose M (g/L): 43.65
Volume V (L): 8.36

t = 58.00 h

Biomass X (g/L): 82.74
Glucose S (g/L): 52.63
Lactic Acid P (g/L): 71.26
Maltose M (g/L): 44.02
Volume V (L): 8.42

t = 59.00 h

Biomass X (g/L): 84.44
Glucose S (g/L): 52.41
Lactic Acid P (g/L): 74.01
Maltose M (g/L): 42.36
Volume V (L): 8.70

t = 60.00 h

Biomass X (g/L): 88.88
Glucose S (g/L): 54.42
Lactic Acid P (g/L): 68.44
Maltose M (g/L): 43.48
Volume V (L): 8.35

t = 61.00 h

Biomass X (g/L): 82.71
Glucose S (g/L): 52.67
Lactic Acid P (g/L): 72.10
Maltose M (g/L): 42.59
Volume V (L): 8.61

t = 62.00 h

Biomass X (g/L): 81.24
Glucose S (g/L): 54.68
Lactic Acid P (g/L): 73.59
Maltose M (g/L): 43.72
Volume V (L): 8.75

t = 63.00 h

Biomass X (g/L): 87.42
Glucose S (g/L): 55.93
Lactic Acid P (g/L): 71.80
Maltose M (g/L): 43.31
Volume V (L): 8.84

t = 64.00 h

Biomass X (g/L): 88.49
Glucose S (g/L): 56.40
Lactic Acid P (g/L): 71.95
Maltose M (g/L): 42.96
Volume V (L): 8.77

t = 65.00 h

Biomass X (g/L): 83.53
Glucose S (g/L): 59.49
Lactic Acid P (g/L): 72.77
Maltose M (g/L): 43.25
Volume V (L): 8.90

t = 66.00 h

Biomass X (g/L): 85.90
Glucose S (g/L): 57.74
Lactic Acid P (g/L): 75.16
Maltose M (g/L): 41.89
Volume V (L): 8.62

t = 67.00 h

Biomass X (g/L): 86.58
Glucose S (g/L): 56.98
Lactic Acid P (g/L): 74.26
Maltose M (g/L): 42.95
Volume V (L): 8.93

t = 68.00 h

Biomass X (g/L): 81.29
Glucose S (g/L): 58.47
Lactic Acid P (g/L): 69.77
Maltose M (g/L): 42.37
Volume V (L): 9.14

t = 69.00 h

Biomass X (g/L): 81.79
Glucose S (g/L): 57.37
Lactic Acid P (g/L): 70.22
Maltose M (g/L): 41.62
Volume V (L): 9.01

t = 70.00 h

Biomass X (g/L): 85.89
Glucose S (g/L): 59.64
Lactic Acid P (g/L): 73.85
Maltose M (g/L): 42.07
Volume V (L): 9.21

t = 71.00 h

Biomass X (g/L): 86.86
Glucose S (g/L): 59.46
Lactic Acid P (g/L): 76.45
Maltose M (g/L): 40.82
Volume V (L): 9.32

t = 72.00 h

Biomass X (g/L): 83.93
Glucose S (g/L): 61.78
Lactic Acid P (g/L): 69.31
Maltose M (g/L): 40.75
Volume V (L): 9.27

t = 73.00 h

Biomass X (g/L): 80.21
Glucose S (g/L): 60.89
Lactic Acid P (g/L): 74.60
Maltose M (g/L): 40.66
Volume V (L): 9.40

t = 74.00 h

Biomass X (g/L): 87.14
Glucose S (g/L): 61.88
Lactic Acid P (g/L): 74.70
Maltose M (g/L): 40.50
Volume V (L): 9.31

t = 75.00 h

Biomass X (g/L): 80.75
Glucose S (g/L): 61.93
Lactic Acid P (g/L): 79.23
Maltose M (g/L): 40.06

Volume V (L): 9.47

Batch 9

Initial Conditions:

Batch: 9.00

X0 (g/L): 5.00

S0 (g/L): 47.25

M0 (g/L): 87.75

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.86

F_vol (L/h): 0.03

S_feed (g/L): 105.00

M_feed (g/L): 195.00

V_final (L): 7.25

Batch 9 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.51
Glucose S (g/L): 47.04
Lactic Acid P (g/L): 5.47
Maltose M (g/L): 88.04
Volume V (L): 4.93

t = 1.00 h

Biomass X (g/L): 7.06
Glucose S (g/L): 46.78
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 87.51
Volume V (L): 5.14

t = 2.00 h

Biomass X (g/L): 7.36
Glucose S (g/L): 48.56
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 85.40
Volume V (L): 5.04

t = 3.00 h

Biomass X (g/L): 11.99
Glucose S (g/L): 48.53
Lactic Acid P (g/L): 0.21
Maltose M (g/L): 83.57
Volume V (L): 5.20

t = 4.00 h

Biomass X (g/L): 16.36
Glucose S (g/L): 44.99
Lactic Acid P (g/L): 8.73
Maltose M (g/L): 84.79
Volume V (L): 5.17

t = 5.00 h

Biomass X (g/L): 13.20
Glucose S (g/L): 43.06

Lactic Acid P (g/L): 5.20
Maltose M (g/L): 83.62
Volume V (L): 5.14

t = 6.00 h

Biomass X (g/L): 25.21
Glucose S (g/L): 41.98
Lactic Acid P (g/L): 11.81
Maltose M (g/L): 80.63
Volume V (L): 5.14

t = 7.00 h

Biomass X (g/L): 25.92
Glucose S (g/L): 38.46
Lactic Acid P (g/L): 16.99
Maltose M (g/L): 82.55
Volume V (L): 5.19

t = 8.00 h

Biomass X (g/L): 32.35
Glucose S (g/L): 35.86
Lactic Acid P (g/L): 19.75
Maltose M (g/L): 81.42
Volume V (L): 5.27

t = 9.00 h

Biomass X (g/L): 38.83
Glucose S (g/L): 33.90
Lactic Acid P (g/L): 23.73
Maltose M (g/L): 79.23
Volume V (L): 5.28

t = 10.00 h

Biomass X (g/L): 36.96
Glucose S (g/L): 28.21
Lactic Acid P (g/L): 26.58
Maltose M (g/L): 78.30
Volume V (L): 5.26

t = 11.00 h

Biomass X (g/L): 42.85

Glucose S (g/L): 28.96
Lactic Acid P (g/L): 32.33
Maltose M (g/L): 78.04
Volume V (L): 5.32

t = 12.00 h

Biomass X (g/L): 47.88
Glucose S (g/L): 25.13
Lactic Acid P (g/L): 30.03
Maltose M (g/L): 75.44
Volume V (L): 5.37

t = 13.00 h

Biomass X (g/L): 50.34
Glucose S (g/L): 22.36
Lactic Acid P (g/L): 33.54
Maltose M (g/L): 75.74
Volume V (L): 5.25

t = 14.00 h

Biomass X (g/L): 59.92
Glucose S (g/L): 20.72
Lactic Acid P (g/L): 35.94
Maltose M (g/L): 72.20
Volume V (L): 5.45

t = 15.00 h

Biomass X (g/L): 56.57
Glucose S (g/L): 19.05
Lactic Acid P (g/L): 42.21
Maltose M (g/L): 75.22
Volume V (L): 5.38

t = 16.00 h

Biomass X (g/L): 60.07
Glucose S (g/L): 15.76
Lactic Acid P (g/L): 44.61
Maltose M (g/L): 71.40
Volume V (L): 5.40

t = 17.00 h

Biomass X (g/L): 64.43
Glucose S (g/L): 17.96
Lactic Acid P (g/L): 45.59
Maltose M (g/L): 74.49
Volume V (L): 5.40

t = 18.00 h

Biomass X (g/L): 63.19
Glucose S (g/L): 15.36
Lactic Acid P (g/L): 49.22
Maltose M (g/L): 71.45
Volume V (L): 5.42

t = 19.00 h

Biomass X (g/L): 67.19
Glucose S (g/L): 14.93
Lactic Acid P (g/L): 45.30
Maltose M (g/L): 70.88
Volume V (L): 5.48

t = 20.00 h

Biomass X (g/L): 70.76
Glucose S (g/L): 14.77
Lactic Acid P (g/L): 53.72
Maltose M (g/L): 69.65
Volume V (L): 5.55

t = 21.00 h

Biomass X (g/L): 70.79
Glucose S (g/L): 12.10
Lactic Acid P (g/L): 54.85
Maltose M (g/L): 69.38
Volume V (L): 5.76

t = 22.00 h

Biomass X (g/L): 72.86
Glucose S (g/L): 13.26
Lactic Acid P (g/L): 54.34
Maltose M (g/L): 70.84
Volume V (L): 5.62

t = 23.00 h

Biomass X (g/L): 73.12
Glucose S (g/L): 12.13
Lactic Acid P (g/L): 58.13
Maltose M (g/L): 65.27
Volume V (L): 5.70

t = 24.00 h

Biomass X (g/L): 71.90
Glucose S (g/L): 13.09
Lactic Acid P (g/L): 51.09
Maltose M (g/L): 69.34
Volume V (L): 5.75

t = 25.00 h

Biomass X (g/L): 77.54
Glucose S (g/L): 9.74
Lactic Acid P (g/L): 54.93
Maltose M (g/L): 67.52
Volume V (L): 5.76

t = 26.00 h

Biomass X (g/L): 82.08
Glucose S (g/L): 11.83
Lactic Acid P (g/L): 55.49
Maltose M (g/L): 64.95
Volume V (L): 5.77

t = 27.00 h

Biomass X (g/L): 78.72
Glucose S (g/L): 11.26
Lactic Acid P (g/L): 59.99
Maltose M (g/L): 64.50
Volume V (L): 5.73

t = 28.00 h

Biomass X (g/L): 80.07
Glucose S (g/L): 12.39
Lactic Acid P (g/L): 56.69
Maltose M (g/L): 67.34
Volume V (L): 5.94

t = 29.00 h

Biomass X (g/L): 82.00
Glucose S (g/L): 12.51
Lactic Acid P (g/L): 61.93
Maltose M (g/L): 63.46
Volume V (L): 5.99

t = 30.00 h

Biomass X (g/L): 77.87
Glucose S (g/L): 11.36
Lactic Acid P (g/L): 64.25
Maltose M (g/L): 62.62
Volume V (L): 5.79

t = 31.00 h

Biomass X (g/L): 85.85
Glucose S (g/L): 11.51
Lactic Acid P (g/L): 63.46
Maltose M (g/L): 64.91
Volume V (L): 5.86

t = 32.00 h

Biomass X (g/L): 82.75
Glucose S (g/L): 12.33
Lactic Acid P (g/L): 63.25
Maltose M (g/L): 63.35
Volume V (L): 5.94

t = 33.00 h

Biomass X (g/L): 86.82
Glucose S (g/L): 11.71
Lactic Acid P (g/L): 60.96
Maltose M (g/L): 61.20
Volume V (L): 5.92

t = 34.00 h

Biomass X (g/L): 81.49
Glucose S (g/L): 13.81
Lactic Acid P (g/L): 62.02
Maltose M (g/L): 62.10

Volume V (L): 6.05

t = 35.00 h

Biomass X (g/L): 88.25

Glucose S (g/L): 12.77

Lactic Acid P (g/L): 67.73

Maltose M (g/L): 60.01

Volume V (L): 6.00

t = 36.00 h

Biomass X (g/L): 86.10

Glucose S (g/L): 10.34

Lactic Acid P (g/L): 67.26

Maltose M (g/L): 62.15

Volume V (L): 6.23

t = 37.00 h

Biomass X (g/L): 85.93

Glucose S (g/L): 13.35

Lactic Acid P (g/L): 65.94

Maltose M (g/L): 61.59

Volume V (L): 6.19

t = 38.00 h

Biomass X (g/L): 85.50

Glucose S (g/L): 12.96

Lactic Acid P (g/L): 65.65

Maltose M (g/L): 60.42

Volume V (L): 6.15

t = 39.00 h

Biomass X (g/L): 84.63

Glucose S (g/L): 12.66

Lactic Acid P (g/L): 62.39

Maltose M (g/L): 59.71

Volume V (L): 6.24

t = 40.00 h

Biomass X (g/L): 88.49

Glucose S (g/L): 13.43

Lactic Acid P (g/L): 65.24

Maltose M (g/L): 60.98

Volume V (L): 6.27

t = 41.00 h

Biomass X (g/L): 86.14

Glucose S (g/L): 15.23

Lactic Acid P (g/L): 72.94

Maltose M (g/L): 58.29

Volume V (L): 6.19

t = 42.00 h

Biomass X (g/L): 81.17

Glucose S (g/L): 16.91

Lactic Acid P (g/L): 65.82

Maltose M (g/L): 56.34

Volume V (L): 6.20

t = 43.00 h

Biomass X (g/L): 83.48

Glucose S (g/L): 16.21

Lactic Acid P (g/L): 69.86

Maltose M (g/L): 57.93

Volume V (L): 6.30

t = 44.00 h

Biomass X (g/L): 85.97

Glucose S (g/L): 15.62

Lactic Acid P (g/L): 72.31

Maltose M (g/L): 59.52

Volume V (L): 6.38

t = 45.00 h

Biomass X (g/L): 87.36

Glucose S (g/L): 17.16

Lactic Acid P (g/L): 67.37

Maltose M (g/L): 55.19

Volume V (L): 6.36

t = 46.00 h

Biomass X (g/L): 86.60

Glucose S (g/L): 16.59

Lactic Acid P (g/L): 71.28

Maltose M (g/L): 55.57

Volume V (L): 6.27

t = 47.00 h

Biomass X (g/L): 89.87

Glucose S (g/L): 16.75

Lactic Acid P (g/L): 73.54

Maltose M (g/L): 56.12

Volume V (L): 6.36

t = 48.00 h

Biomass X (g/L): 82.16

Glucose S (g/L): 16.68

Lactic Acid P (g/L): 71.76

Maltose M (g/L): 55.10

Volume V (L): 6.44

t = 49.00 h

Biomass X (g/L): 87.41

Glucose S (g/L): 16.47

Lactic Acid P (g/L): 79.02

Maltose M (g/L): 53.52

Volume V (L): 6.46

t = 50.00 h

Biomass X (g/L): 82.65

Glucose S (g/L): 18.37

Lactic Acid P (g/L): 68.56

Maltose M (g/L): 55.75

Volume V (L): 6.63

t = 51.00 h

Biomass X (g/L): 89.06

Glucose S (g/L): 19.39

Lactic Acid P (g/L): 75.49

Maltose M (g/L): 56.85

Volume V (L): 6.44

t = 52.00 h

Biomass X (g/L): 87.00

Glucose S (g/L): 19.57
Lactic Acid P (g/L): 72.93
Maltose M (g/L): 53.79
Volume V (L): 6.59

t = 53.00 h

Biomass X (g/L): 88.29
Glucose S (g/L): 19.67
Lactic Acid P (g/L): 75.56
Maltose M (g/L): 52.02
Volume V (L): 6.69

t = 54.00 h

Biomass X (g/L): 85.14
Glucose S (g/L): 19.56
Lactic Acid P (g/L): 75.13
Maltose M (g/L): 50.83
Volume V (L): 6.77

t = 55.00 h

Biomass X (g/L): 90.43
Glucose S (g/L): 21.67
Lactic Acid P (g/L): 72.67
Maltose M (g/L): 52.71
Volume V (L): 6.62

t = 56.00 h

Biomass X (g/L): 89.83
Glucose S (g/L): 20.07
Lactic Acid P (g/L): 75.32
Maltose M (g/L): 53.40
Volume V (L): 6.71

t = 57.00 h

Biomass X (g/L): 88.68
Glucose S (g/L): 22.77
Lactic Acid P (g/L): 70.65
Maltose M (g/L): 52.25
Volume V (L): 6.75

t = 58.00 h

Biomass X (g/L): 86.89
Glucose S (g/L): 21.32
Lactic Acid P (g/L): 76.87
Maltose M (g/L): 51.29
Volume V (L): 6.75

t = 59.00 h

Biomass X (g/L): 88.59
Glucose S (g/L): 22.05
Lactic Acid P (g/L): 76.46
Maltose M (g/L): 50.20
Volume V (L): 6.76

t = 60.00 h

Biomass X (g/L): 85.37
Glucose S (g/L): 21.75
Lactic Acid P (g/L): 77.58
Maltose M (g/L): 49.22
Volume V (L): 6.83

t = 61.00 h

Biomass X (g/L): 89.16
Glucose S (g/L): 23.25
Lactic Acid P (g/L): 78.89
Maltose M (g/L): 51.30
Volume V (L): 6.66

t = 62.00 h

Biomass X (g/L): 91.05
Glucose S (g/L): 25.29
Lactic Acid P (g/L): 77.52
Maltose M (g/L): 51.06
Volume V (L): 6.91

t = 63.00 h

Biomass X (g/L): 89.54
Glucose S (g/L): 25.27
Lactic Acid P (g/L): 77.28
Maltose M (g/L): 47.64
Volume V (L): 6.90

t = 64.00 h

Biomass X (g/L): 85.01
Glucose S (g/L): 26.23
Lactic Acid P (g/L): 75.09
Maltose M (g/L): 51.24
Volume V (L): 6.90

t = 65.00 h

Biomass X (g/L): 87.61
Glucose S (g/L): 26.20
Lactic Acid P (g/L): 75.77
Maltose M (g/L): 49.65
Volume V (L): 7.09

t = 66.00 h

Biomass X (g/L): 83.42
Glucose S (g/L): 24.93
Lactic Acid P (g/L): 80.58
Maltose M (g/L): 50.37
Volume V (L): 6.93

t = 67.00 h

Biomass X (g/L): 87.92
Glucose S (g/L): 26.22
Lactic Acid P (g/L): 81.44
Maltose M (g/L): 46.62
Volume V (L): 6.98

t = 68.00 h

Biomass X (g/L): 87.50
Glucose S (g/L): 26.20
Lactic Acid P (g/L): 77.68
Maltose M (g/L): 46.28
Volume V (L): 7.17

t = 69.00 h

Biomass X (g/L): 81.66
Glucose S (g/L): 28.92
Lactic Acid P (g/L): 78.60
Maltose M (g/L): 47.50
Volume V (L): 7.01

t = 70.00 h

Biomass X (g/L): 83.28
Glucose S (g/L): 26.68
Lactic Acid P (g/L): 76.57
Maltose M (g/L): 48.02
Volume V (L): 7.06

t = 71.00 h

Biomass X (g/L): 85.31
Glucose S (g/L): 29.38
Lactic Acid P (g/L): 81.78
Maltose M (g/L): 49.34
Volume V (L): 7.12

t = 72.00 h

Biomass X (g/L): 85.13
Glucose S (g/L): 29.84
Lactic Acid P (g/L): 80.79
Maltose M (g/L): 47.69
Volume V (L): 7.22

t = 73.00 h

Biomass X (g/L): 82.75
Glucose S (g/L): 31.30
Lactic Acid P (g/L): 84.07
Maltose M (g/L): 46.21
Volume V (L): 7.20

t = 74.00 h

Biomass X (g/L): 86.55
Glucose S (g/L): 29.36
Lactic Acid P (g/L): 77.53
Maltose M (g/L): 44.95
Volume V (L): 7.13

t = 75.00 h

Biomass X (g/L): 86.06
Glucose S (g/L): 31.01
Lactic Acid P (g/L): 78.95
Maltose M (g/L): 47.50

Volume V (L): 7.21

Batch 10

Initial Conditions:

Batch: 10.00

X0 (g/L): 1.00

S0 (g/L): 72.50

M0 (g/L): 72.50

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.00

F_vol (L/h): 0.05

S_feed (g/L): 150.00

M_feed (g/L): 150.00

V_final (L): 8.60

Batch 10 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 73.40
Lactic Acid P (g/L): 0.74
Maltose M (g/L): 73.20
Volume V (L): 4.93

t = 1.00 h

Biomass X (g/L): 1.16
Glucose S (g/L): 74.22
Lactic Acid P (g/L): 0.71
Maltose M (g/L): 71.46
Volume V (L): 5.11

t = 2.00 h

Biomass X (g/L): 2.90
Glucose S (g/L): 75.68
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 71.56
Volume V (L): 5.15

t = 3.00 h

Biomass X (g/L): 3.51
Glucose S (g/L): 76.91
Lactic Acid P (g/L): 1.74
Maltose M (g/L): 71.05
Volume V (L): 5.26

t = 4.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 79.79
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 70.33
Volume V (L): 5.06

t = 5.00 h

Biomass X (g/L): 7.07
Glucose S (g/L): 78.68

Lactic Acid P (g/L): 0.99
Maltose M (g/L): 68.59
Volume V (L): 4.99

t = 6.00 h

Biomass X (g/L): 6.43
Glucose S (g/L): 79.95
Lactic Acid P (g/L): 1.60
Maltose M (g/L): 68.85
Volume V (L): 5.38

t = 7.00 h

Biomass X (g/L): 11.08
Glucose S (g/L): 80.09
Lactic Acid P (g/L): 2.32
Maltose M (g/L): 68.45
Volume V (L): 5.25

t = 8.00 h

Biomass X (g/L): 10.91
Glucose S (g/L): 79.14
Lactic Acid P (g/L): 11.33
Maltose M (g/L): 67.55
Volume V (L): 5.17

t = 9.00 h

Biomass X (g/L): 15.25
Glucose S (g/L): 77.34
Lactic Acid P (g/L): 11.43
Maltose M (g/L): 65.12
Volume V (L): 5.45

t = 10.00 h

Biomass X (g/L): 16.79
Glucose S (g/L): 77.88
Lactic Acid P (g/L): 15.09
Maltose M (g/L): 66.55
Volume V (L): 5.57

t = 11.00 h

Biomass X (g/L): 22.59

Glucose S (g/L): 75.65
Lactic Acid P (g/L): 10.40
Maltose M (g/L): 67.82
Volume V (L): 5.51

t = 12.00 h

Biomass X (g/L): 24.43
Glucose S (g/L): 74.06
Lactic Acid P (g/L): 16.28
Maltose M (g/L): 64.99
Volume V (L): 5.53

t = 13.00 h

Biomass X (g/L): 27.05
Glucose S (g/L): 70.41
Lactic Acid P (g/L): 16.18
Maltose M (g/L): 64.13
Volume V (L): 5.61

t = 14.00 h

Biomass X (g/L): 37.36
Glucose S (g/L): 69.65
Lactic Acid P (g/L): 20.75
Maltose M (g/L): 64.58
Volume V (L): 5.63

t = 15.00 h

Biomass X (g/L): 38.19
Glucose S (g/L): 64.58
Lactic Acid P (g/L): 23.45
Maltose M (g/L): 64.01
Volume V (L): 5.57

t = 16.00 h

Biomass X (g/L): 41.63
Glucose S (g/L): 64.71
Lactic Acid P (g/L): 31.62
Maltose M (g/L): 64.28
Volume V (L): 5.69

t = 17.00 h

Biomass X (g/L): 41.02
Glucose S (g/L): 58.28
Lactic Acid P (g/L): 30.89
Maltose M (g/L): 61.60
Volume V (L): 5.87

t = 18.00 h

Biomass X (g/L): 44.84
Glucose S (g/L): 58.25
Lactic Acid P (g/L): 37.21
Maltose M (g/L): 62.88
Volume V (L): 5.90

t = 19.00 h

Biomass X (g/L): 52.42
Glucose S (g/L): 56.06
Lactic Acid P (g/L): 38.89
Maltose M (g/L): 62.26
Volume V (L): 5.93

t = 20.00 h

Biomass X (g/L): 55.35
Glucose S (g/L): 52.64
Lactic Acid P (g/L): 45.41
Maltose M (g/L): 61.13
Volume V (L): 5.85

t = 21.00 h

Biomass X (g/L): 56.47
Glucose S (g/L): 52.20
Lactic Acid P (g/L): 39.03
Maltose M (g/L): 61.09
Volume V (L): 5.99

t = 22.00 h

Biomass X (g/L): 61.15
Glucose S (g/L): 51.57
Lactic Acid P (g/L): 47.60
Maltose M (g/L): 61.24
Volume V (L): 5.82

t = 23.00 h

Biomass X (g/L): 67.32
Glucose S (g/L): 46.35
Lactic Acid P (g/L): 49.39
Maltose M (g/L): 60.64
Volume V (L): 6.12

t = 24.00 h

Biomass X (g/L): 67.35
Glucose S (g/L): 48.18
Lactic Acid P (g/L): 46.37
Maltose M (g/L): 58.40
Volume V (L): 6.31

t = 25.00 h

Biomass X (g/L): 64.77
Glucose S (g/L): 43.88
Lactic Acid P (g/L): 52.72
Maltose M (g/L): 59.70
Volume V (L): 6.04

t = 26.00 h

Biomass X (g/L): 69.25
Glucose S (g/L): 44.73
Lactic Acid P (g/L): 52.78
Maltose M (g/L): 58.48
Volume V (L): 6.18

t = 27.00 h

Biomass X (g/L): 69.19
Glucose S (g/L): 46.90
Lactic Acid P (g/L): 52.97
Maltose M (g/L): 58.17
Volume V (L): 6.33

t = 28.00 h

Biomass X (g/L): 71.77
Glucose S (g/L): 43.91
Lactic Acid P (g/L): 52.98
Maltose M (g/L): 57.74
Volume V (L): 6.48

t = 29.00 h

Biomass X (g/L): 73.08
Glucose S (g/L): 42.81
Lactic Acid P (g/L): 54.51
Maltose M (g/L): 57.12
Volume V (L): 6.34

t = 30.00 h

Biomass X (g/L): 74.17
Glucose S (g/L): 43.55
Lactic Acid P (g/L): 62.14
Maltose M (g/L): 57.90
Volume V (L): 6.37

t = 31.00 h

Biomass X (g/L): 73.33
Glucose S (g/L): 45.80
Lactic Acid P (g/L): 58.50
Maltose M (g/L): 56.36
Volume V (L): 6.28

t = 32.00 h

Biomass X (g/L): 77.20
Glucose S (g/L): 41.34
Lactic Acid P (g/L): 61.36
Maltose M (g/L): 55.04
Volume V (L): 6.55

t = 33.00 h

Biomass X (g/L): 77.03
Glucose S (g/L): 42.12
Lactic Acid P (g/L): 58.69
Maltose M (g/L): 55.29
Volume V (L): 6.80

t = 34.00 h

Biomass X (g/L): 71.87
Glucose S (g/L): 44.91
Lactic Acid P (g/L): 59.63
Maltose M (g/L): 55.41

Volume V (L): 6.66

t = 35.00 h

Biomass X (g/L): 85.21

Glucose S (g/L): 45.34

Lactic Acid P (g/L): 64.81

Maltose M (g/L): 55.96

Volume V (L): 6.68

t = 36.00 h

Biomass X (g/L): 83.45

Glucose S (g/L): 43.27

Lactic Acid P (g/L): 59.98

Maltose M (g/L): 55.02

Volume V (L): 6.76

t = 37.00 h

Biomass X (g/L): 80.35

Glucose S (g/L): 42.16

Lactic Acid P (g/L): 62.91

Maltose M (g/L): 53.36

Volume V (L): 6.77

t = 38.00 h

Biomass X (g/L): 81.89

Glucose S (g/L): 46.02

Lactic Acid P (g/L): 64.67

Maltose M (g/L): 55.95

Volume V (L): 6.74

t = 39.00 h

Biomass X (g/L): 79.37

Glucose S (g/L): 45.69

Lactic Acid P (g/L): 65.53

Maltose M (g/L): 53.61

Volume V (L): 6.96

t = 40.00 h

Biomass X (g/L): 87.06

Glucose S (g/L): 44.94

Lactic Acid P (g/L): 65.50

Maltose M (g/L): 54.15

Volume V (L): 6.94

t = 41.00 h

Biomass X (g/L): 87.95

Glucose S (g/L): 43.58

Lactic Acid P (g/L): 67.44

Maltose M (g/L): 52.53

Volume V (L): 6.97

t = 42.00 h

Biomass X (g/L): 83.90

Glucose S (g/L): 44.73

Lactic Acid P (g/L): 68.71

Maltose M (g/L): 52.79

Volume V (L): 6.91

t = 43.00 h

Biomass X (g/L): 86.40

Glucose S (g/L): 44.95

Lactic Acid P (g/L): 69.36

Maltose M (g/L): 52.71

Volume V (L): 7.20

t = 44.00 h

Biomass X (g/L): 84.52

Glucose S (g/L): 45.85

Lactic Acid P (g/L): 65.89

Maltose M (g/L): 52.71

Volume V (L): 7.18

t = 45.00 h

Biomass X (g/L): 80.93

Glucose S (g/L): 47.46

Lactic Acid P (g/L): 68.02

Maltose M (g/L): 52.23

Volume V (L): 7.10

t = 46.00 h

Biomass X (g/L): 83.47

Glucose S (g/L): 47.23

Lactic Acid P (g/L): 71.93

Maltose M (g/L): 51.73

Volume V (L): 7.32

t = 47.00 h

Biomass X (g/L): 85.02

Glucose S (g/L): 45.48

Lactic Acid P (g/L): 71.69

Maltose M (g/L): 50.35

Volume V (L): 7.20

t = 48.00 h

Biomass X (g/L): 83.69

Glucose S (g/L): 48.62

Lactic Acid P (g/L): 66.86

Maltose M (g/L): 49.80

Volume V (L): 7.22

t = 49.00 h

Biomass X (g/L): 84.41

Glucose S (g/L): 47.88

Lactic Acid P (g/L): 71.28

Maltose M (g/L): 50.28

Volume V (L): 7.27

t = 50.00 h

Biomass X (g/L): 84.83

Glucose S (g/L): 48.74

Lactic Acid P (g/L): 66.27

Maltose M (g/L): 50.50

Volume V (L): 7.21

t = 51.00 h

Biomass X (g/L): 84.93

Glucose S (g/L): 50.06

Lactic Acid P (g/L): 71.36

Maltose M (g/L): 49.16

Volume V (L): 7.50

t = 52.00 h

Biomass X (g/L): 85.33

Glucose S (g/L): 47.68
Lactic Acid P (g/L): 76.02
Maltose M (g/L): 49.99
Volume V (L): 7.30

t = 53.00 h

Biomass X (g/L): 82.56
Glucose S (g/L): 49.35
Lactic Acid P (g/L): 69.77
Maltose M (g/L): 50.03
Volume V (L): 7.41

t = 54.00 h

Biomass X (g/L): 83.45
Glucose S (g/L): 49.28
Lactic Acid P (g/L): 70.37
Maltose M (g/L): 47.63
Volume V (L): 7.37

t = 55.00 h

Biomass X (g/L): 82.02
Glucose S (g/L): 48.63
Lactic Acid P (g/L): 74.41
Maltose M (g/L): 47.40
Volume V (L): 7.63

t = 56.00 h

Biomass X (g/L): 87.73
Glucose S (g/L): 51.23
Lactic Acid P (g/L): 69.37
Maltose M (g/L): 47.73
Volume V (L): 7.67

t = 57.00 h

Biomass X (g/L): 79.92
Glucose S (g/L): 54.11
Lactic Acid P (g/L): 71.79
Maltose M (g/L): 47.83
Volume V (L): 7.70

t = 58.00 h

Biomass X (g/L): 88.12
Glucose S (g/L): 49.81
Lactic Acid P (g/L): 74.94
Maltose M (g/L): 46.81
Volume V (L): 7.67

t = 59.00 h

Biomass X (g/L): 88.67
Glucose S (g/L): 52.31
Lactic Acid P (g/L): 72.36
Maltose M (g/L): 48.19
Volume V (L): 7.71

t = 60.00 h

Biomass X (g/L): 84.34
Glucose S (g/L): 53.56
Lactic Acid P (g/L): 73.56
Maltose M (g/L): 47.09
Volume V (L): 7.93

t = 61.00 h

Biomass X (g/L): 78.46
Glucose S (g/L): 53.50
Lactic Acid P (g/L): 76.10
Maltose M (g/L): 46.47
Volume V (L): 7.87

t = 62.00 h

Biomass X (g/L): 84.05
Glucose S (g/L): 53.74
Lactic Acid P (g/L): 75.61
Maltose M (g/L): 44.88
Volume V (L): 8.01

t = 63.00 h

Biomass X (g/L): 84.71
Glucose S (g/L): 52.95
Lactic Acid P (g/L): 75.50
Maltose M (g/L): 44.72
Volume V (L): 7.93

t = 64.00 h

Biomass X (g/L): 82.22
Glucose S (g/L): 55.15
Lactic Acid P (g/L): 74.33
Maltose M (g/L): 45.87
Volume V (L): 8.09

t = 65.00 h

Biomass X (g/L): 83.37
Glucose S (g/L): 54.56
Lactic Acid P (g/L): 76.21
Maltose M (g/L): 43.46
Volume V (L): 8.26

t = 66.00 h

Biomass X (g/L): 88.43
Glucose S (g/L): 53.49
Lactic Acid P (g/L): 75.29
Maltose M (g/L): 46.00
Volume V (L): 8.23

t = 67.00 h

Biomass X (g/L): 86.13
Glucose S (g/L): 57.74
Lactic Acid P (g/L): 76.86
Maltose M (g/L): 44.70
Volume V (L): 8.17

t = 68.00 h

Biomass X (g/L): 81.99
Glucose S (g/L): 57.99
Lactic Acid P (g/L): 74.86
Maltose M (g/L): 47.47
Volume V (L): 8.29

t = 69.00 h

Biomass X (g/L): 80.15
Glucose S (g/L): 57.50
Lactic Acid P (g/L): 75.79
Maltose M (g/L): 46.38
Volume V (L): 8.29

t = 70.00 h

Biomass X (g/L): 83.23
Glucose S (g/L): 58.20
Lactic Acid P (g/L): 74.12
Maltose M (g/L): 43.92
Volume V (L): 8.36

t = 71.00 h

Biomass X (g/L): 83.38
Glucose S (g/L): 59.64
Lactic Acid P (g/L): 76.53
Maltose M (g/L): 43.74
Volume V (L): 8.37

t = 72.00 h

Biomass X (g/L): 85.02
Glucose S (g/L): 59.21
Lactic Acid P (g/L): 74.92
Maltose M (g/L): 44.48
Volume V (L): 8.44

t = 73.00 h

Biomass X (g/L): 82.27
Glucose S (g/L): 60.46
Lactic Acid P (g/L): 74.40
Maltose M (g/L): 44.86
Volume V (L): 8.53

t = 74.00 h

Biomass X (g/L): 84.76
Glucose S (g/L): 59.64
Lactic Acid P (g/L): 77.77
Maltose M (g/L): 41.66
Volume V (L): 8.44

t = 75.00 h

Biomass X (g/L): 88.20
Glucose S (g/L): 61.91
Lactic Acid P (g/L): 77.84
Maltose M (g/L): 41.29

Volume V (L): 8.49